

Exam : JN0-281

**Title : Data Center, Associate
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Vendor : Juniper

Version : V12.65

QUESTION NO: 1

Which static routing parameter will silently drop the packet if it is set as the next hop?

- A. Reject
- B. Resolve
- C. Readvertise
- D. Discard

Answer: D

Explanation:

When the discard option is configured as the next hop for a static route, it silently drops any packets that match the route without sending any notification to the sender.

Step-by-Step Breakdown:

Discard Behavior:

If a route uses the discard next hop, the router drops the packet without generating any ICMP message or error back to the sender. This is useful for creating null routes to prevent routing loops or blackhole traffic intentionally.

Reject vs. Discard:

The reject next hop, in contrast, drops the packet but sends an ICMP Destination Unreachable message back to the source.

Juniper Reference:

Static Route Behavior: In Junos, the discard option ensures packets matching a static route are dropped silently, providing a way to discard traffic without alerting the source.

QUESTION NO: 2

In the Junos OS, which feature is used to create an alternate next hop with a unique preference for a static route?

- A. Preference
- B. Resolve
- C. Next-hop
- D. Qualified-next-hop

Answer: D

Explanation:

In Junos OS, the qualified-next-hop feature is used to specify an alternate next hop for a static route, along with a unique preference value.

Step-by-Step Breakdown:

Qualified-Next-Hop:

A qualified-next-hop allows you to define multiple next hops for a static route, each with its own preference. This provides flexibility by allowing the router to choose the best available next hop based on reachability and preference.

Use Case:

If the primary next hop becomes unreachable, the router can automatically switch to the alternate next hop defined by the qualified-next-hop with a higher preference value.

Command Example:

```
set routing-options static route 10.10.10.0/24 qualified-next-hop 192.168.1.1 preference 5 set routing-options static route 10.10.10.0/24 qualified-next-hop 192.168.1.2 preference 10
```

Preference:

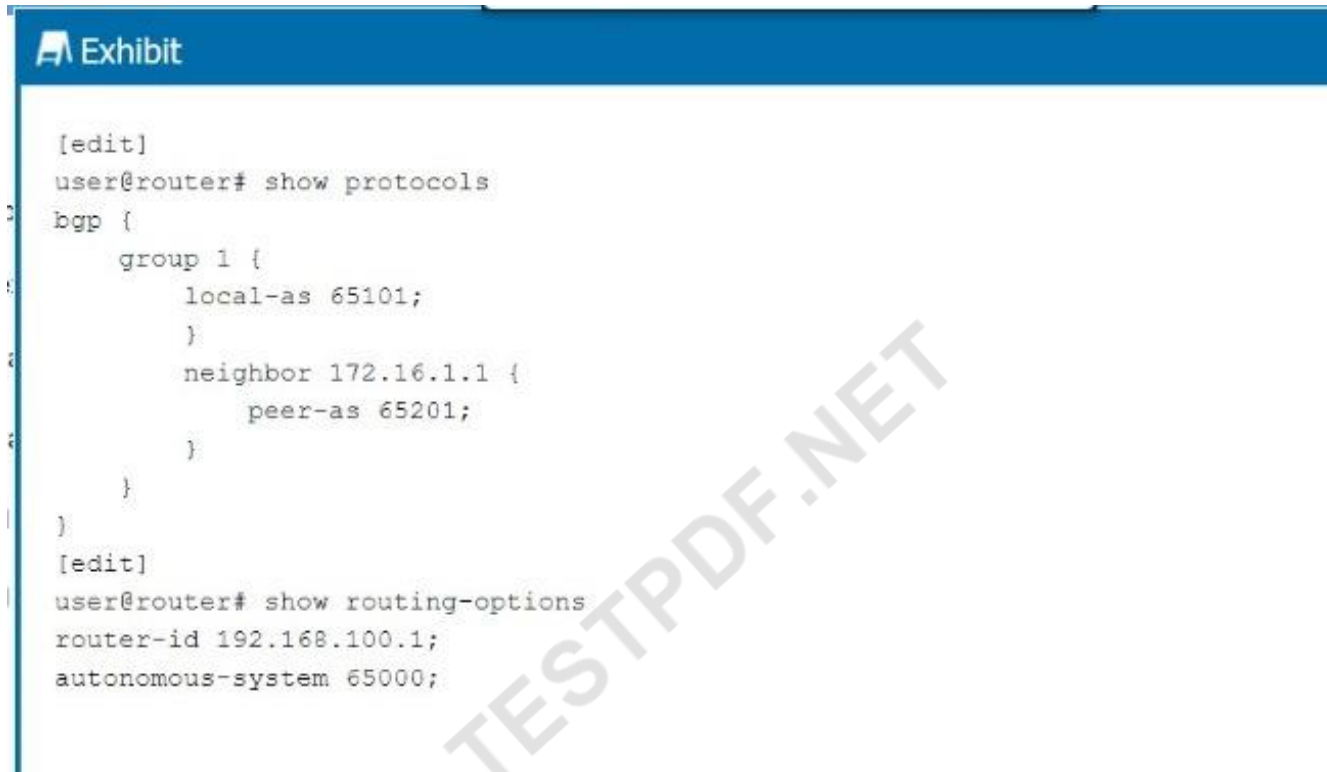
The next hop with the lowest preference is chosen first. If it becomes unavailable, the router will use the higher preference next hop.

Juniper Reference:

Qualified-Next-Hop: This feature is used to configure backup or alternate next hops for static routes in Juniper devices.

QUESTION NO: 3

Exhibit:



```
[edit]
user@router# show protocols
bgp {
  group 1 {
    local-as 65101;
  }
  neighbor 172.16.1.1 {
    peer-as 65201;
  }
}
[edit]
user@router# show routing-options
router-id 192.168.100.1;
autonomous-system 65000;
```

The exhibit shows a Juniper router configuration. The BGP configuration includes a group named '1' with a local AS of 65101, and a neighbor at 172.16.1.1 with a peer AS of 65201. The routing options show the router ID as 192.168.100.1 and the autonomous system as 65000. A large diagonal watermark 'TESTPDF.NET' is visible across the configuration text.

Referring to the exhibit, which statement is correct?

- A. The configuration will commit successfully and BGP group 1 will operate as IBGP.
- B. The configuration will commit successfully and BGP group 1 will operate as EBGP.
- C. BGP group 1 requires a type external parameter.
- D. BGP group 1 requires a type internal parameter.

Answer: B

Explanation:

In the exhibit, BGP is configured with local AS 65101 and a neighbor at 172.16.1.1 in peer AS 65201. This setup involves two different Autonomous Systems (AS), indicating an External BGP (EBGP) configuration.

Step-by-Step Breakdown:

EBGP vs. IBGP:

EBGP is used between routers in different ASes. In this case, the local AS is 65101 and the peer AS is 65201, meaning the BGP session is EBGP.

IBGP is used between routers within the same AS, which is not applicable here as the AS numbers are different.

BGP Group Configuration:

The configuration does not require a type external parameter because Junos OS automatically recognizes the session as EBGP when the local and peer AS numbers are different.

The BGP session will operate as EBGP, and the configuration will commit successfully.

Juniper Reference:

BGP Configuration: In Juniper, EBGP is automatically recognized when the local and peer AS numbers differ, without needing to specify type external.

QUESTION NO: 4

Which two statements about IBGP are correct? (Choose two.)

- A. By default, IBGP has a TTL of 1.
- B. IBGP uses AS path for loop prevention.
- C. By default, IBGP has a TTL of 255.
- D. IBGP uses full mesh for loop prevention.

Answer: C,D

Explanation:

IBGP (Internal Border Gateway Protocol) is used to exchange routing information between routers within the same AS (Autonomous System).

Step-by-Step Breakdown:

TTL of 255:

By default, IBGP sessions are established with a TTL (Time to Live) value of 255. This allows IBGP neighbors to communicate over multiple hops within the AS without requiring any additional configuration.

Full Mesh Requirement:

IBGP requires a logical full mesh between all IBGP routers to ensure that routing information is fully distributed within the AS. Since IBGP does not propagate routes learned from one IBGP peer to another by default, a full mesh topology is needed unless route reflectors or BGP confederations are used.

Juniper Reference:

IBGP Full Mesh: Juniper recommends using route reflectors in large networks to simplify IBGP full-mesh requirements.

QUESTION NO: 5

Which statement is correct about areas in OSPF?

- A. An OSPF area is used to segment Layer 2 broadcast domains.
- B. OSPF areas are used to isolate the effects of a broadcast storm.
- C. OSPF areas are used to reduce the size of the link-state database.
- D. An OSPF area is used to signify the autonomous system to which each device belongs.

Answer: C

Explanation:

In OSPF (Open Shortest Path First), areas are used to segment a network into smaller, more manageable pieces to improve scalability. By dividing a network into areas, OSPF can reduce the size of the link-state database (LSDB), which helps routers process updates more efficiently.

Step-by-Step Breakdown:

Purpose of OSPF Areas:

OSPF areas allow for hierarchical routing within the OSPF domain. Routers in the same area have identical LSDBs, but routers in different areas do not exchange full link-state information. Instead, they exchange summarized routes, which reduces the LSDB size and CPU/memory usage.

Benefits:

Reducing the LSDB size improves scalability and ensures faster convergence in larger networks. Area 0 is the backbone area, and all other areas must connect to it, forming a hierarchical structure.

Juniper Reference:

OSPF Configuration: Areas in OSPF are configured to optimize network performance by limiting the scope of link-state advertisements (LSAs) to within an area.

QUESTION NO: 6

Which state in the adjacency process do OSPF routers check the MTU size?

- A. Init
- B. Exchange
- C. Done
- D. ExStart

Answer: B

Explanation:

In OSPF, routers exchange link-state information in different stages to establish full adjacency. The MTU size is checked during the Exchange state.

Step-by-Step Breakdown:**OSPF Adjacency Process:**

OSPF routers go through multiple stages when forming an adjacency: Down, Init, 2-Way, ExStart, Exchange, Loading, and Full.

Exchange State:

During the Exchange state, OSPF routers exchange Database Description (DBD) packets to describe their link-state databases. The MTU size is checked at this stage to ensure both routers can successfully exchange these packets without fragmentation.

If there is an MTU mismatch, the routers may fail to proceed past the Exchange state.

Juniper Reference:

MTU Checking in OSPF: Junos uses the Exchange state to check for MTU mismatches, ensuring that routers can properly exchange database information without packet fragmentation issues.

QUESTION NO: 7

When troubleshooting an OSPF neighborship, you notice that the router stopped at the ExStart state. What is the cause of this result?

- A. The priority is set to 255.
- B. There is an interval timing mismatch.
- C. There is an area ID mismatch.
- D. There is an MTU mismatch.

Answer: D

Explanation:

When an OSPF (Open Shortest Path First) neighborship is stuck in the ExStart state, it usually points to a mismatch in Maximum Transmission Unit (MTU) settings between two routers trying to establish the adjacency. The ExStart state is where OSPF routers negotiate the master-slave relationship and exchange DBD (Database Description) packets.

Step-by-Step Breakdown:

OSPF Neighbor States: OSPF goes through several states to establish an adjacency with a neighbor:

Down: No hello packets have been received.

Init: Hello packets are received, but bidirectional communication isn't confirmed.

2-Way: Bidirectional communication is established.

ExStart: The routers are negotiating who will be the master and who will be the slave, and begin to exchange DBD packets.

Exchange: The routers start exchanging the database information.

Loading: The routers process the Link-State Advertisements (LSAs).

Full: The adjacency is fully established.

MTU Mismatch Issue:

During the ExStart state, both OSPF routers must agree on their MTU values. If there is an MTU mismatch between the two routers, OSPF neighbors will fail to move from the ExStart to the Exchange state. The router with the larger MTU setting will not accept DBD packets from the router with a smaller MTU because the packets may exceed the smaller MTU size.

In Juniper devices, this behavior can be identified by examining the MTU settings using the show interfaces command and ensuring both routers have matching MTU configurations. To resolve this issue, either match the MTU settings on both routers or configure OSPF to ignore MTU mismatches using the command set protocols ospf ignore-mtu.

Juniper Reference:

Junos Command: show ospf neighbor helps diagnose neighbor states.

MTU Adjustment: set interfaces <interface-name> mtu <size> can be used to set the MTU values correctly.

QUESTION NO: 8

Which three actions are required to implement filter-based forwarding? (Choose three.)

- A. You must create an instance-type forwarding routing instance.
- B. You must create an instance-type vrf routing instance.
- C. You must create a match filter.
- D. You must create a security policy.
- E. You must create a RIB group.

Answer: A,C,E

Explanation:

Filter-Based Forwarding (FBF) in Junos OS allows traffic to be routed based on specific criteria such as source address, rather than just the destination address. This is useful in scenarios like policy routing or providing multiple paths for different types of traffic.

Step-by-Step Breakdown:**Instance-Type Forwarding:**

You must create an instance-type forwarding routing instance. This routing instance allows

for different routing tables based on the incoming packet filter.

Command:

```
set routing-instances FBF-instance instance-type forwarding
```

Match Filter:

You need to create a filter to match the traffic that will be forwarded according to your custom routing policy. This filter is applied to an interface to determine which traffic will use the custom forwarding instance.

Command Example:

```
set firewall family inet filter FBF-filter term 1 from source-address <address> set firewall family inet filter FBF-filter term 1 then routing-instance FBF-instance RIB Group:
```

A RIB (Routing Information Base) group is necessary to share routes between the primary routing table and the custom routing instance. This allows FBF traffic to use the routing information from other routing tables.

Command Example:

```
set routing-options rib-groups FBF-group import-rib inet.0  
set routing-instances FBF-instance routing-options rib-group FBF-group
```

Juniper Reference:

FBF Configuration: Filter-based forwarding requires these specific steps to redirect traffic to a custom routing table based on filter criteria.

QUESTION NO: 9

Which two statements are correct about rules for EBGp and IBGP? (Choose two.)

- A. EBGp peers have a TTL of 1, while IBGP peers have a TTL of 255.
- B. EBGp peers have a TTL of 255, while IBGP peers have a TTL of 1.
- C. EBGp routes are more preferred than IBGP routes.
- D. IBGP routes are more preferred than EBGp routes.

Answer: A,C

Explanation:

EBGP (External BGP) and IBGP (Internal BGP) operate with different rules due to the nature of their relationships.

Step-by-Step Breakdown:

TTL Differences:

EBGP: By default, EBGp peers have a TTL of 1, meaning they must be directly connected, or the TTL needs to be manually increased for multihop EBGp.

IBGP: IBGP peers within the same AS have a TTL of 255, as they are expected to communicate over multiple hops within the AS.

Preference for EBGp Routes:

Routes learned via EBGp are typically preferred over IBGP routes. This is because EBGp routes are considered more reliable since they originate outside the AS, while IBGP routes are internal.

Juniper Reference:

BGP Configuration: The different handling of TTL and route preferences between EBGp and IBGP ensures proper route selection and security within Junos-based networks.

QUESTION NO: 10

Which statement about switches is correct?

- A. Each port is a member of VLAN 2 by default.
- B. Every port is in a unique collision domain.
- C. Each port is in a unique broadcast domain by default.
- D. All ports reside in the same collision domain.

Answer: B

Explanation:

Each port on a modern switch creates a separate collision domain. This allows multiple devices to communicate simultaneously without collisions on different ports.

Step-by-Step Breakdown:

Collision Domain:

A collision domain is a network segment where data packets can collide if two devices send packets simultaneously.

On a switch, each port creates a separate collision domain, so collisions only occur if two devices connected to the same port (through a hub, for instance) try to send data at the same time.

Switches vs Hubs:

Unlike hubs, which have one large collision domain, switches isolate collisions to individual ports, improving performance.

Juniper Reference:

Switch Port Behavior: In Juniper switches, each port operates in its own collision domain, enhancing network efficiency by reducing the chances of packet collisions.

QUESTION NO: 11

You are creating an IP fabric underlay and want to use OSPF as your routing protocol. In this scenario, which statement is correct?

- A. All leaf devices must be configured in separate OSPF areas.
- B. All leaf and spine devices must be the same model to ensure the proper load-balancing behavior.
- C. Interface speeds should be the same throughout the fabric to ensure that all links are utilized.
- D. All spine devices must use the same router ID.

Answer: C

Explanation:

When creating an IP fabric underlay using OSPF as the routing protocol, consistent interface speeds are important to ensure optimal traffic distribution and utilization of all links.

Step-by-Step Breakdown:

OSPF and Interface Speeds:

OSPF calculates the cost of a link based on its bandwidth. The default cost calculation in OSPF is:

$$\text{Cost} = \frac{\text{Reference Bandwidth}}{\text{Interface Bandwidth}}$$

If interface speeds vary significantly, OSPF may choose paths with lower cost (higher

bandwidth), resulting in some links being underutilized.

Equal Utilization:

To ensure that all links are equally utilized in an IP fabric, it is recommended to maintain uniform interface speeds across the fabric. This ensures balanced load sharing across all available paths.

Juniper Reference:

IP Fabric with OSPF: Juniper recommends consistent interface speeds to maintain even traffic distribution and optimal link utilization in IP fabric underlay designs.

QUESTION NO: 12

What is the behavior of the default export policy for OSPF?

- A. Accept all routes.
- B. Reject all routes.
- C. Redistribute all routes.
- D. Forward all routes.

Answer: B

Explanation:

In Junos, the default export policy for OSPF is to reject all routes from being exported.

Step-by-Step Breakdown:

Default Export Policy:

By default, OSPF in Junos does not export any routes to other routing protocols or neighbors. This is a safety mechanism to prevent unintended route advertisements.

Custom Export Policies:

If you need to export routes, you must create a custom export policy that explicitly defines which routes to advertise.

Example: You can create an export policy to redistribute static or connected routes into OSPF.

Juniper Reference:

OSPF Export Behavior: In Juniper devices, the default policy for OSPF is to reject route advertisements unless explicitly configured otherwise through custom policies.

QUESTION NO: 13

Referring to the exhibit, why are the BGP routes hidden?

- A. Load balancing is not enabled.
- B. There are too many hops to the destination.
- C. The BGP next hop is unreachable.
- D. Other routes are selected because of better metrics.

Answer: C

Explanation:

In the exhibit, the BGP routes are marked as hidden. This typically happens when the routes are not considered valid for use, but they remain in the routing table for reference. One common reason for BGP routes being hidden is that the next hop for these routes is unreachable.

Step-by-Step Breakdown:

BGP Next Hop:

In BGP, when a route is received from a neighbor, the next hop is the IP address that must be reachable for the route to be used. If the next hop is unreachable (i.e., the router cannot find a path to the next-hop IP), the route is marked as hidden.

Analyzing the Exhibit:

The exhibit shows that the BGP next hop for all hidden routes is 10.4.4.4. If this IP is unreachable, the BGP routes from that neighbor will not be considered valid, even though they appear in the routing table.

Verification:

Use the command `show route 10.4.4.4` to check if the next-hop IP is reachable.

If the next-hop is not reachable, the BGP routes will be hidden. Resolving the next-hop reachability issue (e.g., fixing an IGP route or an interface) will allow the BGP routes to become active.

Juniper Reference:

Junos Command: `show route hidden` displays routes that are not considered for forwarding.

Troubleshooting: Check the next hop reachability for hidden BGP routes using `show route <next-hop>`.

QUESTION NO: 14

Which two statements are correct about EVPN-VXLAN overlay networking? (Choose two.)

- A. It is the only option to provide reachability between servers that reside in the same network segment in a data center.
- B. BGP provides the control plane within the overlay network.
- C. An encapsulation of the original packet is required to transport the packet across the network.
- D. OSPF provides the control plane within the overlay network.

Answer: B,C

Explanation:

EVPN-VXLAN is an overlay technology used in data center networks to extend Layer 2 services over a Layer 3 network.

Step-by-Step Breakdown:

BGP Control Plane:

BGP (Border Gateway Protocol) is used as the control plane for EVPN-VXLAN. BGP advertises MAC addresses and IP address reachability information across the VXLAN network, enabling efficient multi-tenant Layer 2 connectivity over a Layer 3 infrastructure.

Encapsulation:

VXLAN (Virtual Extensible LAN) encapsulates Layer 2 frames into Layer 3 packets. This encapsulation allows Layer 2 traffic to be transported across a Layer 3 network, effectively creating a tunnel for Ethernet frames.

Juniper Reference:

EVPN-VXLAN Configuration: Juniper supports EVPN-VXLAN with BGP as the control plane, allowing scalable Layer 2 connectivity over a routed infrastructure in modern data centers.

QUESTION NO: 15

Which two statements are correct about aggregate routes and generated routes? (Choose two.)

- A. An aggregate route does not have a forwarding next hop.
- B. An aggregate route has a forwarding next hop.
- C. A generated route has a forwarding next hop.
- D. A generated route does not have a forwarding next hop.

Answer: A,C

Explanation:

Aggregate routes and generated routes are used to create summarized routes in Junos, but they behave differently in terms of forwarding.

Step-by-Step Breakdown:

Aggregate Routes:

An aggregate route summarizes a set of more specific routes, but it does not have a direct forwarding next hop. Instead, it points to the more specific routes for actual packet forwarding.

Generated Routes:

A generated route also summarizes specific routes, but it has a forwarding next hop that is determined based on the availability of contributing routes. The generated route can be used to directly forward traffic.

Juniper Reference:

Aggregate and Generated Routes: In Junos, aggregate routes rely on more specific routes for forwarding, while generated routes can forward traffic directly based on their next-hop information.

QUESTION NO: 16

What are two requirements for an IP fabric? (Choose two.)

- A. a Layer 3 routing protocol
- B. a single connection between each spine and leaf
- C. a single connection between each leaf
- D. a Layer 2 switching protocol

Answer: A,B

Explanation:

An IP fabric is a network architecture commonly used in data centers to provide scalable, high-throughput connectivity using a spine-leaf topology.

Step-by-Step Breakdown:

Layer 3 Routing Protocol:

An IP fabric relies on a Layer 3 routing protocol, typically BGP or OSPF, to provide routing between the leaf and spine switches. This ensures efficient traffic forwarding across the network.

Single Connection Between Spine and Leaf:

In an IP fabric, each leaf switch connects to every spine switch with a single connection. This ensures that traffic between any two leaf switches can travel through the spine layer in just two hops.

Juniper Reference:

Spine-Leaf Design: Juniper's IP fabric implementations are designed for scalability and low-latency routing, often using protocols like BGP for Layer 3 control.

QUESTION NO: 17

You want to enable a Junos device to support aggregated Ethernet interfaces. In this scenario, which configuration hierarchy would you use?

- A. [edit switch-options]
- B. [edit system]
- C. [edit interfaces]
- D. [edit chassis]

Answer: D

Explanation:

To configure aggregated Ethernet (AE) interfaces on a Junos device, the configuration is done under the [edit chassis] hierarchy.

Step-by-Step Breakdown:

Chassis Configuration:

The chassis configuration is responsible for enabling the hardware to support Link Aggregation Groups (LAGs), allowing multiple physical interfaces to be bundled into a single logical interface for load balancing and redundancy.

Command Example:

```
set chassis aggregated-devices ethernet device-count <number>
```

This command enables a specific number of aggregated Ethernet interfaces on the device.

Juniper Reference:

LAG Configuration in Junos: The chassis hierarchy is used to allocate and manage hardware resources for aggregated Ethernet interfaces in Juniper devices.

QUESTION NO: 18

What are three correct layer names used in legacy hierarchical network design? (Choose three.)

- A. Access layer
- B. Modular layer
- C. Aggregation layer
- D. Core layer
- E. Function layer

Answer: A,C,D

Explanation:

In legacy hierarchical network design, three key layers are used to create a scalable and structured network:

Step-by-Step Breakdown:

Access Layer:

The access layer is where end devices, such as computers and IP phones, connect to the network. It typically involves switches that provide connectivity for devices at the edge of the network.

Aggregation Layer (Distribution Layer):

The aggregation layer (also called the distribution layer) aggregates traffic from multiple access layer devices and applies policies such as filtering and QoS. It also provides redundancy and load balancing.

Core Layer:

The core layer provides high-speed connectivity between aggregation layer devices and facilitates traffic within the data center or between different network segments.

Juniper Reference:

Legacy Hierarchical Design: Juniper networks often follow the traditional three-layer design (Access, Aggregation, and Core) to ensure scalability and high performance.

QUESTION NO: 19

What is the main purpose of Bidirectional Forwarding Detection (BFD)?

- A.** to detect network path failures
- B.** to determine if the forwarding routes are correct
- C.** to detect the forwarding protocol
- D.** to determine packet round-trip latency

Answer: A

Explanation:

Bidirectional Forwarding Detection (BFD) is a network protocol used to detect failures in the network path between two devices quickly.

Step-by-Step Breakdown:**Path Failure Detection:**

BFD provides a low-overhead mechanism for detecting failures in forwarding paths across Layer 3 networks. It is much faster than traditional routing protocol timers and can detect failures within milliseconds.

BFD in Routing:

BFD can be integrated with routing protocols like OSPF, BGP, or IS-IS to trigger a faster convergence when a network path goes down.

Juniper Reference:

BFD Configuration: Juniper devices use BFD to monitor network paths and ensure fast failure detection, enhancing network resilience.

QUESTION NO: 20

Which statement is correct about an IRB interface?

- A.** An IRB interface switches traffic within the same VLAN.
- B.** An IRB interface trunks together VLANs on different switches.
- C.** An IRB interface is a physical Layer 3 interface that connects VLANs together.
- D.** An IRB interface is a Layer 3 interface that can be used to route between VLANs.

Answer: D

Explanation:

An IRB (Integrated Routing and Bridging) interface provides routing functionality between VLANs at Layer 3, allowing devices in different VLANs to communicate with each other.

Step-by-Step Breakdown:**IRB Functionality:**

The IRB interface enables routing between different VLANs by acting as a Layer 3 gateway. Traffic within the same VLAN is handled by Layer 2 switching, while traffic between VLANs is routed through the IRB interface.

Layer 3 Routing Between VLANs:

Each VLAN can be assigned an IP address on the IRB interface, which allows traffic to flow between VLANs based on Layer 3 IP routing.

Juniper Reference:

IRB Interface Configuration: Juniper supports IRB for inter-VLAN routing on devices like the EX and QFX series switches, facilitating Layer 3 communication in data centers.

QUESTION NO: 21

Which statement is correct about a three-stage IP fabric underlay?

- A. Every ingress interface into the fabric is only two hops away from the egress interface.
- B. Every spine device can communicate directly with other spine devices.
- C. Every leaf device can communicate directly with other leaf devices.
- D. Every server that connects to a three-stage IP fabric must be multihomed.

Answer: A

Explanation:

In a three-stage IP fabric (also known as a Clos fabric), traffic between any two points (ingress to egress) in the fabric is only two hops away.

Step-by-Step Breakdown:

Three-Stage IP Fabric:

Leaf Layer: Leaf switches connect directly to servers and edge devices.

Spine Layer: Spine switches provide connectivity between leaf switches but do not connect to each other directly.

Two-Hop Communication:

In this architecture, every leaf switch is connected to every spine switch. Therefore, when a packet enters the fabric via an ingress leaf switch, it is forwarded to a spine switch, which then directs the packet to the correct egress leaf switch. This path always involves exactly two hops:

Ingress leaf → Spine → Egress leaf.

Benefits:

This consistent two-hop path ensures predictable latency and makes the network highly scalable while maintaining low complexity.

Juniper Reference:

IP Fabric Architecture: This two-hop property of Clos fabrics is a hallmark of spine-leaf designs, as supported by Juniper's QFX and EX switches in data centers.

QUESTION NO: 22

Exhibit:

Exhibit

```
{master:0}[edit switch-options]
user@switch# show
interface ge-0/0/1.0 {
    persistent-learning;
}
```

Referring to the exhibit, which behavior does this configuration enable on the ge-0/0/1.0 interface?

- A.** This configuration enables a MAC address learned on the interface to be persistently retained in the Ethernet-switching table, even after a reboot.
- B.** This configuration enables the device to place a MAC address that persistently causes network errors into a special protected VLAN.
- C.** This configuration enables the device to shut down the interface when a particular MAC address persistently sends broadcast traffic.
- D.** This configuration enables the interface to learn and remember MAC addresses, until the device is rebooted.

Answer: A

Explanation:

The configuration in the exhibit shows the persistent-learning feature enabled on interface ge-0/0/1.0.

Step-by-Step Breakdown:

Persistent Learning:

Persistent-learning ensures that the MAC addresses learned on the interface are retained in the Ethernet-switching table, even after a device reboot. This prevents the need to re-learn MAC addresses after the device restarts, improving stability and reducing downtime.

Use Case:

This feature is particularly useful in environments where the re-learning of MAC addresses could cause temporary disruptions or delays in communication, such as in critical Layer 2 network segments.

Command Example:

set switch-options interface ge-0/0/1.0 persistent-learning

Juniper Reference:

Persistent MAC Learning: In Junos, enabling persistent-learning ensures that learned MAC addresses are not lost during reboots, contributing to smoother network operations in environments where stability is crucial.

QUESTION NO: 23

Which three technologies improve high availability and convergence in a data center network? (Choose three.)

- A. graceful restart (GR)
- B. Bidirectional Forwarding Detection (BFD)
- C. link loss adjacency
- D. Failover Group (FG)
- E. link aggregation group (LAG)

Answer: A,B,E

Explanation:

High availability and fast convergence are critical in data center networks to minimize downtime and maintain optimal performance. The following technologies contribute to achieving these goals:

Graceful Restart (GR):

GR allows routers to maintain forwarding state during control plane restarts, ensuring continuous packet forwarding while minimizing network disruptions.

Bidirectional Forwarding Detection (BFD):

BFD provides fast detection of path failures, allowing routing protocols to converge quickly by detecting link failures much faster than traditional timers.

Link Aggregation Group (LAG):

LAG increases both redundancy and bandwidth by combining multiple physical links into one logical link, providing load balancing and fault tolerance.

Juniper Reference:

High Availability Techniques: These technologies are fundamental in ensuring rapid recovery and failover within Juniper-based data center environments.

QUESTION NO: 24

What is the primary purpose of an IRB Layer 3 interface?

- A. to provide load balancing
- B. to provide a default VLAN ID
- C. to provide inter-VLAN routing
- D. to provide port security

Answer: C

Explanation:

The primary purpose of an IRB (Integrated Routing and Bridging) interface is to enable inter-VLAN routing in a Layer 3 environment. An IRB interface in Junos combines the functionality of both Layer 2 bridging (switching) and Layer 3 routing, allowing devices in different VLANs to communicate with each other.

Step-by-Step Breakdown:**VLANs and Layer 2 Switching:**

Devices within the same VLAN can communicate directly through Layer 2 switching.

However, communication between devices in different VLANs requires Layer 3 routing.

IRB Interface for Inter-VLAN Routing:

The IRB interface provides a Layer 3 gateway for each VLAN, enabling routing between VLANs. Without an IRB interface, devices in different VLANs would not be able to communicate.

Configuration:

In Juniper devices, the IRB interface is configured by assigning Layer 3 IP addresses to it.

These IP addresses serve as the default gateway for devices in different VLANs.

Example configuration:

```
set interfaces irb unit 0 family inet address 192.168.1.1/24
```

```
set vlans vlan-10 l3-interface irb.0
```

This allows VLAN 10 to use the IRB interface for routing.

Juniper Reference:

IRB Use Case: Inter-VLAN routing is essential in data centers where multiple VLANs are deployed, and Juniper's EX and QFX series switches support IRB configurations for this purpose.

QUESTION NO: 25

Referring to the exhibit, you notice that after committing the configuration, the ae0 and ae1 interfaces appear in a link down state.

 **Exhibit**

```
[edit]
user@switch# show chassis
aggregated-devices {
  ethernet {
    device-count 2;
  }
}
[edit]
user@switch# run show interfaces terse | match ae
ae0          up    down
ae1          up    down
```

Which statement is correct in this scenario?

- A.** No operational interfaces have been added to the LAG interfaces.
- B.** No traffic is traversing the LAG interfaces.

- C. The LAG interfaces are in a passive state.
- D. The LAG interfaces are in aggressive mode.

Answer: A

Explanation:

In the exhibit, the ae0 and ae1 interfaces are in a link down state. This occurs when no physical interfaces (member interfaces) have been added to the LAG (Link Aggregation Group) interfaces, or the member interfaces are not operational.

Step-by-Step Breakdown:

LAG Configuration:

A LAG interface (aggregated Ethernet interface) is a logical interface that combines multiple physical interfaces for redundancy and increased bandwidth. The LAG will only be operational if at least one member interface is active and configured correctly.

No Operational Member Interfaces:

If no member interfaces are added or if the member interfaces are down, the LAG will remain in a down state, as shown in the exhibit for ae0 and ae1.

Resolution:

Verify that physical interfaces have been added to the LAG using commands like:

LAG Interface Status: In Juniper, the link status of the LAG depends on its member interfaces, which must be operational for the LAG to function.

QUESTION NO: 26

Which two statements are correct about VLAN tags? (Choose two.)

- A. VLAN tags carry a VLAN ID and priority.
- B. VLAN tags are required on access ports.
- C. VLAN tags require multiple forwarding tables.
- D. VLAN tags can be inserted or removed by trunk interfaces.

Answer: A,D

Explanation:

VLAN tags are used in Ethernet frames to identify and differentiate traffic between multiple VLANs. They are especially important for devices like switches that handle multiple VLANs on the same physical link.

Step-by-Step Breakdown:

VLAN Tag Contents:

VLAN ID: The tag contains a 12-bit VLAN ID field that identifies the VLAN to which the frame belongs.

Priority: The tag also includes a 3-bit priority field (also known as 802.1p priority) used for QoS (Quality of Service) to prioritize traffic.

Trunk Ports and VLAN Tagging:

Trunk Ports are used to carry traffic for multiple VLANs across a single link. These interfaces insert (tag) VLAN identifiers into frames when they leave the switch and remove (untag) them when frames enter the switch.

Access Ports:

VLAN tags are typically not used on access ports (ports that connect to end devices) since those ports are configured to be part of a single VLAN, and the traffic doesn't need VLAN tags.

Juniper Reference:

VLAN Tagging: Juniper switches support VLAN tagging and ensure that frames are tagged or untagged as they traverse trunk or access ports, respectively.

QUESTION NO: 27

When considering bidirectional forwarding detection, which two statements are correct? (Choose two.)

- A. The BFD default minimum interval is 3.
- B. You can configure BFD per interface within the protocol stanza.
- C. The BFD operation always consists of minimum intervals and multipliers.
- D. The BFD default multiplier is 5.

Answer: B,C

Explanation:

Bidirectional Forwarding Detection (BFD) is a protocol used to detect faults in the forwarding path between two routers. It provides rapid failure detection, enhancing the performance of routing protocols like OSPF, BGP, and IS-IS.

Step-by-Step Breakdown:

Per Interface Configuration:

BFD can be configured on a per-interface basis within the protocol stanza (e.g., OSPF, BGP). This allows granular control over where BFD is enabled and the failure detection intervals for specific interfaces.

Minimum Interval and Multiplier:

BFD uses a minimum interval (the time between BFD control packets) and a multiplier (the number of missed packets before the path is declared down). The combination of these two defines the detection time for failures.

Juniper Reference:

BFD Configuration: In Juniper, BFD is configurable within routing protocol stanzas, with the failure detection mechanism always based on minimum intervals and multipliers.

QUESTION NO: 28

When using spine and leaf fabric architectures, what is the role of each device? (Choose two.)

- A. Spine nodes are used for host connectivity.
- B. Spine nodes are used for transit to other leaf nodes.
- C. Leaf nodes are used for traffic to other leafs.
- D. Leaf nodes are used for host connectivity.

Answer: B,D

Explanation:

In a spine-leaf fabric architecture, which is commonly used in data center designs, each device has a distinct role to ensure efficient and scalable network traffic flow.

Step-by-Step Breakdown:

Spine Nodes:

The spine nodes form the backbone of the fabric and are responsible for transit traffic between leaf nodes. They connect to every leaf switch and provide multiple paths for traffic between leaf nodes, ensuring redundancy and load balancing.

Leaf Nodes:

The leaf nodes are used for host connectivity. These switches connect to servers, storage, or edge routers. They also connect to the spine switches to reach other leaf switches.

Juniper Reference:

Spine-Leaf Architecture: In Juniper's IP fabric designs, spine switches handle inter-leaf communication, while leaf switches manage host and endpoint connectivity.

QUESTION NO: 29

What information in the Ethernet header is used to populate the bridging table?

- A.** destination address
- B.** source address
- C.** type
- D.** protocol

Answer: B

Explanation:

The source MAC address in the Ethernet header is used to populate the bridging table (also called the MAC address table) on a switch. When a frame arrives at a switch, the switch examines the source MAC address and records it along with the ingress port in its MAC address table.

Step-by-Step Breakdown:**Learning Process:**

When an Ethernet frame arrives on a switch port, the switch looks at the source MAC address and adds this MAC address to the MAC table along with the port it was received on. This process is called MAC learning.

Purpose:

The switch uses this information to determine the correct port to send frames destined for that MAC address in future transmissions, thus ensuring efficient Layer 2 forwarding.

Juniper Reference:

Ethernet Switching: Juniper switches use source MAC addresses to build and maintain the MAC address table, which is essential for Layer 2 switching.

QUESTION NO: 30**Exhibit:**

 Exhibit

```
[edit routing-options]
user@Router# show
static {
    route 0.0.0.0/0 {
        next-hop 172.25.11.254;
        qualified-next-hop 172.25.11.200 {
            preference 140;
        }
    }
}
```

Referring to the exhibit, what is the route preference of the 172.25.11.254 next hop?

- A. 5
- B. 10
- C. 130
- D. 140

Answer: A

Explanation:

In the exhibit, we see two next-hop addresses for the default static route (0.0.0.0/0):

The first next hop is 172.25.11.254, with no specified preference.

The second next hop is 172.25.11.200, with a specified preference of 140.

Step-by-Step Breakdown:

Default Static Route Preference:

If no preference is explicitly set for a next hop in Junos, it defaults to 5 for static routes.

Determining Preference:

In this case, the next hop 172.25.11.254 does not have an explicit preference defined, so it will use the default value of 5. The second next hop has a preference of 140, which is higher, meaning it will only be used if the primary next hop is unavailable.

Juniper Reference:

Static Route Preference: In Junos, the default preference for static routes is 5, and this value is applied unless overridden by the preference parameter.

QUESTION NO: 31

How does OSPF calculate the best path to a particular prefix?

- A. It finds the path with the numerically lowest cost.
- B. It finds the path with the shortest autonomous system path.
- C. It finds the path with the least number of hops.

D. It finds the path with the numerically lowest route preference.

Answer: A

Explanation:

OSPF (Open Shortest Path First) calculates the best path based on the cost of the route, which is derived from the bandwidth of the interfaces along the path.

Step-by-Step Breakdown:

OSPF Path Selection:

OSPF assigns a cost to each link, typically based on the link's bandwidth (higher bandwidth equals lower cost).

The OSPF algorithm computes the shortest path to a destination by adding the costs of all links in the path. The path with the numerically lowest total cost is chosen as the best path.

Cost Calculation:

The OSPF cost can be manually adjusted or automatically calculated using the default formula:

$$\text{Cost} = \frac{\text{Reference Bandwidth}}{\text{Link Bandwidth}}$$

Reference:

OSPF Best Path Selection: OSPF selects the path with the lowest cumulative cost, ensuring efficient use of higher-bandwidth links in Junos networks.

QUESTION NO: 32

What are two reasons why you would deploy an IP fabric instead of a traditional Layer 2 network in a data center? (Choose two.)

- A.** Layer 2 networks only support a single broadcast domain.
- B.** IP fabrics are better suited to smaller networks where scale is less important.
- C.** Layer 3 networks support load balancing.
- D.** Layer 2 networks are susceptible to loops.

Answer: C,D

Explanation:

IP fabrics are Layer 3-centric network designs often used in data centers due to their scalability, efficient routing, and loop-free architecture.

Step-by-Step Breakdown:

Layer 3 Load Balancing:

IP fabrics use Equal-Cost Multipath (ECMP) to distribute traffic across multiple paths, providing effective load balancing and improving bandwidth utilization. This capability is absent in traditional Layer 2 networks, which do not support ECMP for routing decisions.

Layer 2 Loops:

Layer 2 networks are prone to loops because of the lack of TTL (Time-to-Live) mechanisms. Spanning Tree Protocol (STP) is required to prevent loops, but it can introduce inefficiencies by blocking links. In contrast, IP fabrics based on Layer 3 protocols are loop-free and do not need STP.

Juniper Reference:

IP Fabric: Juniper's IP fabric solutions offer efficient Layer 3 routing with built-in load balancing and loop prevention, making them ideal for modern data center architectures.

QUESTION NO: 33

Which signaling protocol is used for EVPN?

- A. OSPF
- B. PIM
- C. IS-IS
- D. BGP

Answer: D

Explanation:

EVPN (Ethernet Virtual Private Network) is a standard protocol used for building Layer 2 and Layer 3 VPNs over an IP or MPLS network. The signaling protocol used for EVPN is BGP (Border Gateway Protocol).

Step-by-Step Breakdown:

BGP as the EVPN Signaling Protocol:

EVPN uses BGP to exchange MAC address reachability information between routers (PE devices). This enables devices to learn which MAC addresses are reachable through which PE devices, facilitating Layer 2 forwarding across an IP or MPLS core.

BGP Extensions for EVPN:

BGP is extended with new address families (e.g., EVPN NLRI) to carry both MAC and IP address information, allowing for scalable and efficient multi-tenant network solutions.

Juniper Reference:

Junos EVPN Configuration: Juniper uses BGP as the control plane for EVPN to exchange MAC and IP route information between different data center devices.

QUESTION NO: 34

Within your router, you want to verify that you are learning routes from a remote BGP peer at IP address 10.10.100.1. Which command would satisfy the requirement?

- A. show route receive-protocol bgp 10.10.100.1
- B. show route protocol bgp table inet.0 10.10.100.1
- C. show route advertise-protocol bgp 10.10.100.1
- D. show route protocol bgp source-gateway 10.10.100.1

Answer: A

Explanation:

To verify that your router is learning routes from a remote BGP peer at a specific IP address (e.g., 10.10.100.1), the correct command to use is show route receive-protocol bgp.

Step-by-Step Breakdown:

BGP Route Learning:

The show route receive-protocol bgp command displays the routes that have been received from a specified BGP peer. This helps in confirming that the remote peer is sending routes correctly and that your router is receiving them.

Command Example:

```
show route receive-protocol bgp 10.10.100.1
```

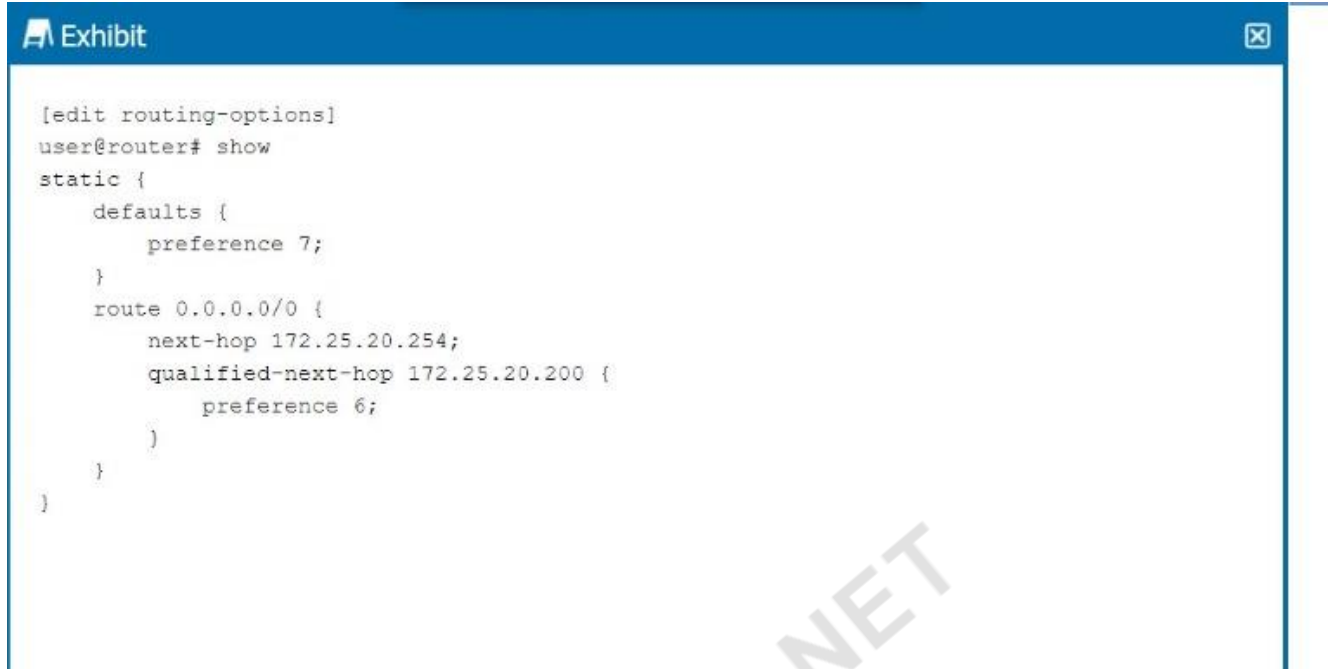
This will show all routes that have been received from the BGP peer with IP address 10.10.100.1.

Juniper Reference:

BGP Route Verification: Use this command to troubleshoot and verify that routes from a specific BGP peer are being received.

QUESTION NO: 35

Exhibit:



```
[edit routing-options]
user@router# show
static {
    defaults {
        preference 7;
    }
    route 0.0.0.0/0 {
        next-hop 172.25.20.254;
        qualified-next-hop 172.25.20.200 {
            preference 6;
        }
    }
}
```

Referring to the exhibit, which next hop will be preferred in the routing table?

- A. Next hop IP address 172.25.20.254 will be preferred.
- B. Neither next hop will be preferred.
- C. Next hop IP address 172.25.20.200 will be preferred.
- D. Both next hops will be preferred.

Answer: C

Explanation:

In the exhibit, we see a static route configuration with two possible next hops for the default route (0.0.0.0/0):

next-hop 172.25.20.254 with the default preference of 7.

qualified-next-hop 172.25.20.200 with a preference of 6.

Step-by-Step Breakdown:

Preference Value:

In Junos OS, the preference value is used to determine which route should be preferred in the routing table. The lower the preference value, the higher the priority for the route.

Comparison:

In this case:

The next hop 172.25.20.254 has a preference of 7.

The qualified-next-hop 172.25.20.200 has a preference of 6.

Preferred Next Hop:

Since 172.25.20.200 has a lower preference (6) compared to 172.25.20.254 (7), it will be the preferred next hop in the routing table, assuming both next hops are reachable.

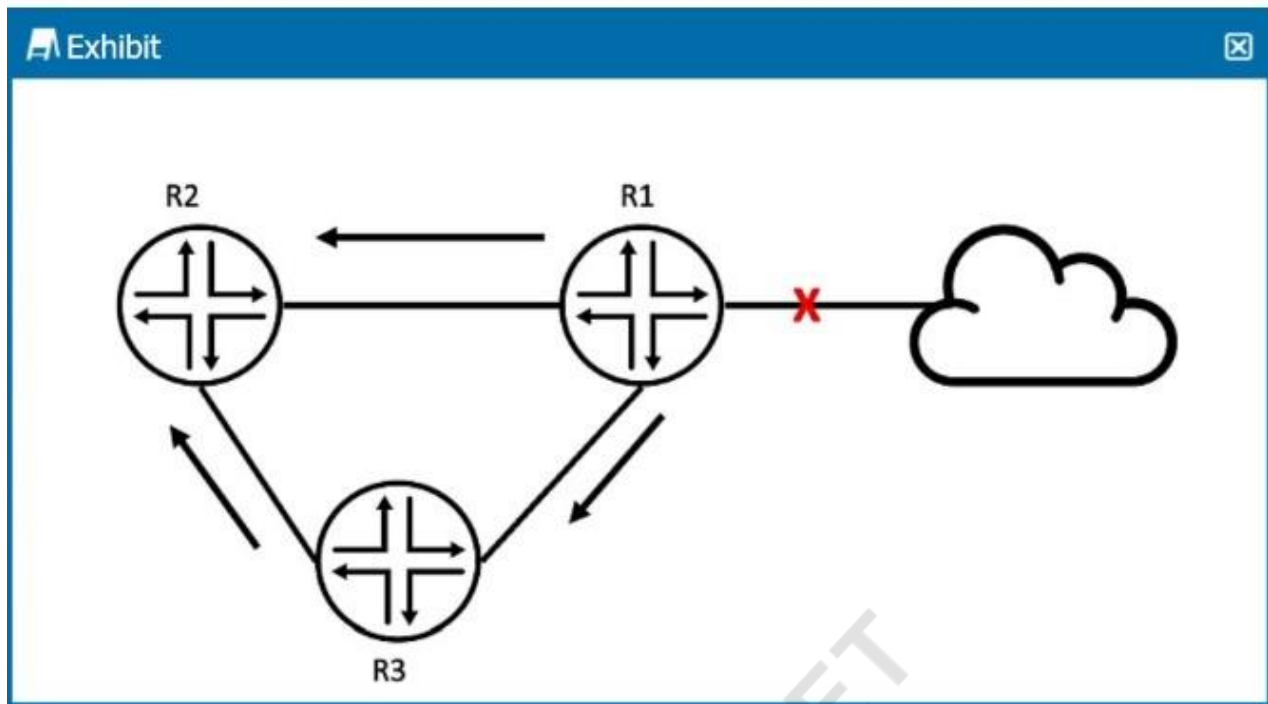
Juniper Reference:

Qualified Next Hop: In Junos, static routes with multiple next-hop options are selected based

on the preference value, with the lower value being preferred.

QUESTION NO: 36

Exhibit:



R2 received an OSPF update from R1, and it received the same update from R3. Referring to the exhibit, what will R2 do?

- A. R2 ignores the update from R1.
- B. R2 does nothing with R3's update.
- C. R2 ignores the update from R3.
- D. R2 acknowledges R3 and discards it.

Answer: C

Explanation:

In the exhibit, R2 receives the same OSPF update from both R1 and R3. OSPF has mechanisms to prevent unnecessary processing of duplicate LSAs (Link-State Advertisements).

Step-by-Step Breakdown:

OSPF LSA Processing:

OSPF uses LSAs to exchange link-state information between routers. When a router receives an LSA, it checks if it already has a copy of the LSA in its Link-State Database (LSDB).

Duplicate LSAs:

If R2 has already received and processed the update from R1, it will ignore the update from R3 because it already has the same LSA in its database. OSPF uses the concept of flooding, but it does not reprocess LSAs that it already knows about.

R2 Behavior:

R2 will keep the update from R1 (the first one it received) and will ignore the same LSA from R3, as it is already in the LSDB.

Juniper Reference:

OSPF LSA Processing: Junos adheres to OSPF standards, ensuring that duplicate LSAs are not processed multiple times to avoid unnecessary recalculations.

QUESTION NO: 37

What is the default route preference of a static route in the Junos OS?

- A. 0
- B. 10
- C. 1
- D. 5

Answer: D

Explanation:

In Junos OS, the default route preference for a static route is 5. Route preference values are used to determine which route should be installed in the routing table when multiple routes to the same destination are available.

Step-by-Step Breakdown:

Static Route Preference:

A static route, by default, has a preference of 5, making it a highly preferred route. Lower preference values are more preferred in Junos, meaning static routes take precedence over most dynamic routing protocol routes, such as OSPF (preference 10) or BGP (preference 170).

Route Preference:

Route preference is a key factor in the Junos routing decision process. Routes with lower preference values are preferred and installed in the forwarding table.

Juniper Reference:

Static Routes: In Junos, the default preference for static routes is 5, making them more preferred than most dynamic routes.

QUESTION NO: 38

A routing policy has been created to advertise OSPF routes in BGP. Which statement is correct in this scenario?

- A. Apply the policy as an export policy within BGP.
- B. Apply the policy as an export policy within OSPF.
- C. Apply the policy as an import policy within BGP.
- D. Apply the policy as an import policy within OSPF.

Answer: A

Explanation:

When advertising OSPF routes into BGP, the appropriate routing policy should be applied as an export policy in BGP.

Step-by-Step Breakdown:

OSPF to BGP Route Advertisement:

Routes learned via OSPF (a dynamic IGP) need to be exported into BGP to be advertised to external BGP peers. In Junos OS, this is done using export policies.

Export Policies in BGP:

An export policy controls which routes are advertised out of a BGP session. In this scenario, the routing policy must be applied to BGP as an export policy to export the OSPF-learned

routes to external BGP peers.

Policy Configuration:

Example configuration:

```
set policy-options policy-statement EXPORT_OSPF term 1 from protocol ospf set policy-  
options policy-statement EXPORT_OSPF term 1 then accept set protocols bgp group  
<group-name> export EXPORT_OSPF
```

This policy ensures that only OSPF routes are exported into BGP.

Juniper Reference:

Routing Policy: Export policies are used in BGP to control route advertisements to peers, including those learned via OSPF.

QUESTION NO: 39

Leaf and spine data centers are used to better accommodate which type of traffic?

- A. north-east
- B. east-west
- C. north-west
- D. south-east

Answer: B

Explanation:

In modern data centers, the shift toward leaf-spine architectures is driven by the need to handle increased east-west traffic, which is traffic between servers within the same data center. Unlike traditional hierarchical data center designs, where most traffic was "north-south" (between users and servers), modern applications often involve server-to-server communication (east-west) to enable services like distributed databases, microservices, and virtualized workloads.

Leaf-Spine Architecture:

Leaf Layer: This layer consists of switches that connect directly to servers or end-host devices. These switches serve as the access layer.

Spine Layer: The spine layer comprises high-performance switches that provide interconnectivity between leaf switches. Each leaf switch connects to every spine switch, creating a non-blocking fabric that optimizes traffic flow within the data center.

East-West Traffic Accommodation:

In traditional three-tier architectures (core, aggregation, access), traffic had to traverse multiple layers, leading to bottlenecks when servers communicated with each other. Leaf-spine architectures address this by creating multiple equal-cost paths between leaf switches and the spine. Since each leaf switch connects directly to every spine switch, the architecture facilitates quick, low-latency communication between servers, which is essential for east-west traffic flows.

Juniper's Role:

Juniper Networks provides a range of solutions that optimize for east-west traffic in a leaf-spine architecture, notably through:

QFX Series Switches: Juniper's QFX series switches are designed for the leaf and spine architecture, delivering high throughput, low latency, and scalability to accommodate the traffic demands of modern data centers.

EVPN-VXLAN: Juniper uses EVPN-VXLAN to create a scalable Layer 2 and Layer 3 overlay

network across the data center. This overlay helps enhance east-west traffic performance by enabling network segmentation and workload mobility across the entire fabric.

Key Features That Support East-West Traffic:

Equal-Cost Multipath (ECMP): ECMP enables the use of multiple paths between leaf and spine switches, balancing the traffic and preventing any one path from becoming a bottleneck. This is crucial in handling the high volume of east-west traffic.

Low Latency: Spine switches are typically high-performance devices that minimize the delay between leaf switches, which improves the efficiency of server-to-server communications.

Scalability: As the demand for east-west traffic grows, adding more leaf and spine switches is straightforward, maintaining consistent performance without redesigning the entire network.

In summary, the leaf-spine architecture is primarily designed to handle the increase in east-west traffic within data centers, and Juniper provides robust solutions to enable this architecture through its switch platforms and software solutions like EVPN-VXLAN.

QUESTION NO: 40

A switch receives a frame with a MAC address of FF-FF-FF-FF-FF-FF. Which action will the switch take on this frame?

- A.** It will flood it out of all interfaces, except for the ingress interface.
- B.** It will flood it out of all interfaces, except for the directly connected VLAN.
- C.** It will flood it out of all interfaces, except for the next-hop interface.
- D.** It will flood it out of all interfaces.

Answer: A

Explanation:

A MAC address of FF-FF-FF-FF-FF-FF is the Ethernet broadcast address. When a switch receives a frame with this destination MAC address, it is required to forward the frame to all interfaces except the one it was received on.

Step-by-Step Breakdown:

Broadcast Frame Handling:

When a frame with the broadcast MAC address is received, the switch will flood it out of all active ports that belong to the same VLAN as the incoming frame. The broadcast frame is not sent back out of the ingress interface (the interface where the frame was originally received).

Purpose of Flooding:

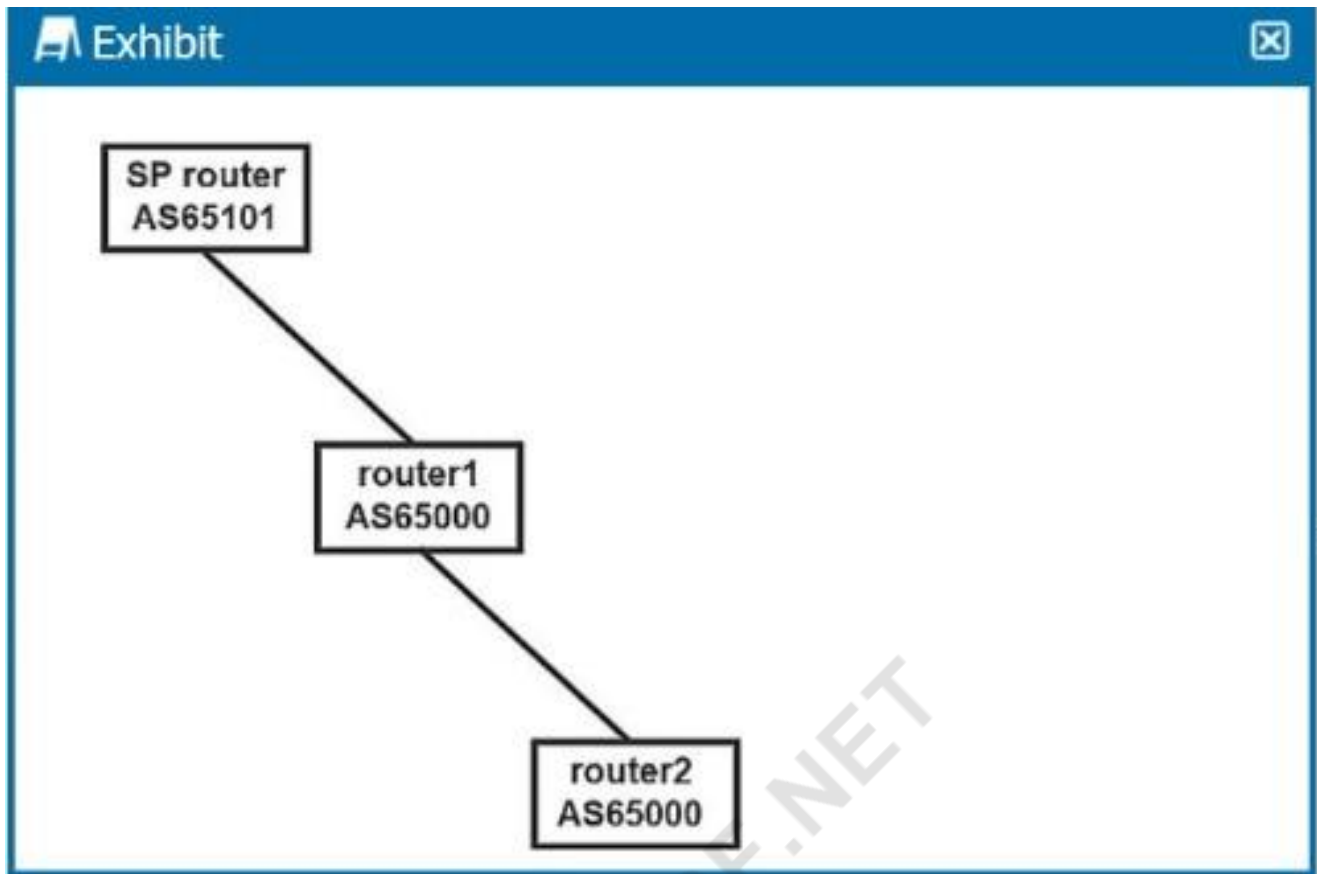
Broadcasting is used to ensure that the frame reaches all devices within the broadcast domain (all devices within the same VLAN), which may not have a specific entry for the MAC address in their MAC address table.

Juniper Reference:

Layer 2 Frame Forwarding: Juniper switches flood broadcast frames to all ports in the same VLAN, except the port the frame was received on.

QUESTION NO: 41

Exhibit:



Referring to the exhibit, which two statements are correct about default BGP advertisements? (Choose two.)

- A. When routes advertised by router2 are received by the SP router, they will contain the next-hop address of router2.
- B. When routes advertised by router2 are received by the SP router, they will contain the next-hop address of router1.
- C. When routes advertised by the SP router are received by router2, they will contain the next-hop address of the SP router.
- D. When routes advertised by the SP router are received by router2, they will contain the next-hop address of router1.

Answer: B,D

Explanation:

The exhibit shows a BGP peering scenario between three routers: router1 and router2 are part of the same AS (AS65000), while the SP router is in a different AS (AS65101). This indicates an EBGP (External BGP) peering between the SP router and router1, and IBGP between router1 and router2.

Step-by-Step Breakdown:

Next-Hop Behavior in BGP:

IBGP: In IBGP, the next-hop address is not modified when advertising routes within the same AS. Thus, when router1 advertises routes learned from router2 to the SP router, it will keep the next-hop address of router1, not router2.

EBGP: In EBGP, the next-hop address is modified. When router1 receives routes from the SP router, it will advertise them to router2 with the next-hop address of router1.

Route Propagation:

Routes received by router1 from router2 will be advertised to the SP router with router1 as the next hop.

Similarly, routes advertised by the SP router will be passed on to router2, with router1 remaining as the next hop.

Juniper Reference:

BGP Next-Hop: Juniper's BGP implementations follow standard BGP next-hop behavior, where the next-hop is modified in EBGP but not in IBGP, ensuring proper route advertisement across autonomous systems.

QUESTION NO: 42

Which Junos OS routing table stores IPv6 addresses?

- A. inet.0
- B. inet0.6
- C. inet.6
- D. inet6.0

Answer: D

Explanation:

In Junos OS, routing information is stored in different routing tables depending on the protocol and address family. For IPv6 addresses, the routing table used is inet6.0.

Step-by-Step Explanation:

Routing Tables in Junos:

inet.0: This is the primary routing table for IPv4 unicast routes.

inet6.0: This is the primary routing table for IPv6 unicast routes.

inet.3: This routing table is used for MPLS-related routing.

Other routing tables, like inet.1, inet.2, are used for multicast and other specific purposes.

inet6.0 Routing Table:

When IPv6 is enabled on a Juniper router, all the IPv6 routes are stored in the inet6.0 table. This includes both direct routes (connected networks) and learned routes (from dynamic routing protocols like OSPFv3, BGP, etc.).

Verification:

To view IPv6 routes, the command show route table inet6.0 is used. This will display the contents of the IPv6 routing table, showing the network prefixes, next-hop addresses, and protocol information for each route.

Juniper Reference:

Junos Command: Use show route table inet6.0 to check IPv6 routing entries.

IPv6 Routing: Ensure that the IPv6 protocol is enabled on interfaces and that routing protocols like OSPFv3 or BGP are properly configured for IPv6 traffic handling.

QUESTION NO: 43

What are two device roles in a five-member Virtual Chassis? (Choose two.)

- A. PFE
- B. Control-board
- C. Line card
- D. Routing-engine

Answer: C,D

Explanation:

In a Virtual Chassis (VC) configuration, multiple Juniper switches are interconnected to form a single logical device. Each member switch in the Virtual Chassis plays a specific role.

Step-by-Step Breakdown:**Line Card Role:**

Member switches acting as line cards provide additional ports for traffic forwarding but do not perform control or routing functions. These switches depend on the routing engine to handle control-plane tasks.

Routing Engine Role:

A switch in the routing-engine role is responsible for control-plane operations such as routing protocol management and control of the Virtual Chassis.

Virtual Chassis Roles:

Master Routing Engine: Handles control-plane functions and manages the entire Virtual Chassis.

Backup Routing Engine: Takes over if the master fails.

Line Card: Provides additional ports and handles data-plane operations.

Juniper Reference:

Virtual Chassis: In a five-member Virtual Chassis, multiple switches act as line cards, while one or more switches are designated as the routing engines (master and backup).

QUESTION NO: 44

What are two consequences of having all network devices in a single collision domain? (Choose two.)

- A.** The amount of network resource consumption does not change.
- B.** The chance of packet collision is decreased.
- C.** The chance of packet collision is increased.
- D.** The amount of network resource consumption is increased.

Answer: C,D

Explanation:

A collision domain is a network segment where data packets can "collide" with one another when being sent on the same network medium.

Step-by-Step Breakdown:**Increased Collision Probability:**

If all devices are in a single collision domain, the likelihood of packet collisions increases as more devices attempt to send packets simultaneously, leading to network inefficiencies.

Increased Resource Consumption:

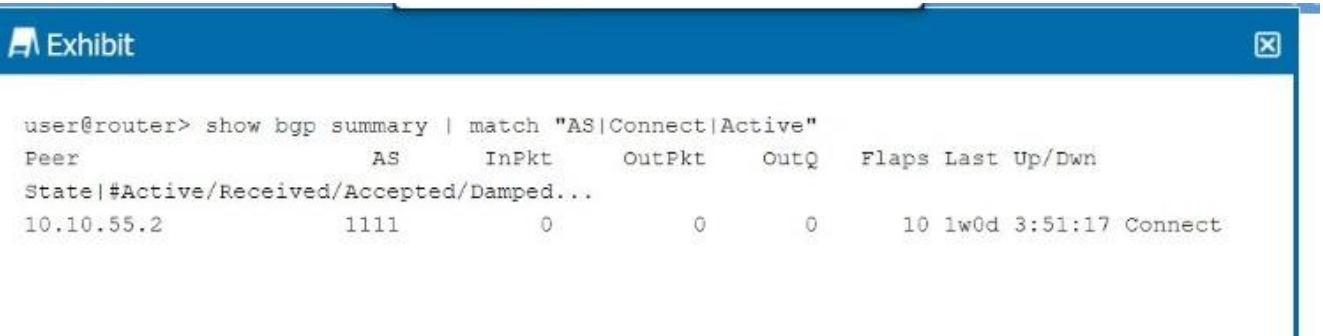
More collisions result in increased network resource consumption as devices need to retransmit packets, causing higher utilization of bandwidth and slowing down network performance.

Juniper Reference:

Collision Domains: Proper network segmentation using switches reduces collision domains, thereby improving network performance and reducing packet collisions.

QUESTION NO: 45

You are troubleshooting a downed BGP session.



```

user@router> show bgp summary | match "AS|Connect|Active"
Peer          AS      InPkt    OutPkt    OutQ    Flaps Last Up/Dwn
State|#Active/Received/Accepted/Damped...
10.10.55.2      1111         0         0         0      10 1w0d 3:51:17 Connect

```

Referring to the exhibit, what is the cause of the problem?

- A. The UDP session between the peers has not been established.
- B. The local peer has sent an Open message but not received one from the remote peer.
- C. The TCP session between the peers has not been established.
- D. The local peer has sent an Update message but not received one from the remote peer.

Answer: C

Explanation:

The BGP session in the exhibit shows the state as Connect, which indicates that the TCP session between the BGP peers has not been fully established.

Step-by-Step Breakdown:

BGP State "Connect":

The Connect state is the second stage in the BGP finite state machine (FSM). At this stage, BGP is trying to establish a TCP session with the peer, but the session has not yet been successfully established.

A successful TCP three-way handshake (SYN, SYN-ACK, ACK) is required before BGP can progress to the OpenSent state, where the peers exchange BGP Open messages.

Possible Causes:

A firewall blocking TCP port 179.

Incorrect IP addresses or network connectivity issues between the BGP peers.

Juniper Reference:

BGP Troubleshooting: In Junos, if a BGP session is stuck in the Connect state, the issue is likely due to a failure in establishing the underlying TCP connection.

QUESTION NO: 46

What is the definition of a trunk interface on a switch?

- A. An interface that carries multiple VLANs.
- B. An interface that carries high bandwidth.
- C. An interface that connects directly to powerful servers.
- D. An interface that carries excess traffic.

Answer: A

Explanation:

A trunk interface on a switch is used to carry traffic for multiple VLANs between switches or between a switch and another network device, like a router. Trunk interfaces use 802.1Q tagging to identify which VLAN the traffic belongs to.

Step-by-Step Breakdown:

Trunk Ports:

Trunk ports are typically used for inter-switch links or switch-to-router links where multiple VLANs need to be carried over the same physical connection.

VLAN traffic is tagged with a VLAN ID to ensure that it is properly identified as it crosses the trunk link.

802.1Q VLAN Tagging:

Trunk ports use 802.1Q to tag Ethernet frames with the VLAN ID. This ensures that frames are correctly forwarded to the appropriate VLANs on the other side of the trunk.

Juniper Reference:

Trunk Interface Configuration: In Juniper switches, trunk ports are configured to carry tagged traffic for multiple VLANs, which is essential for interconnecting multiple network segments.

QUESTION NO: 47

A generated route is configured under which hierarchy?

- A. [edit policy-options]
- B. [edit routing-instance]
- C. [edit routing-options]
- D. [edit protocols]

Answer: C

Explanation:

A generated route in Junos OS is configured under the [edit routing-options] hierarchy.

Step-by-Step Breakdown:

Generated Routes:

A generated route is created based on the presence of more specific routes in the routing table. It acts as a summary route and is generated when any of its contributing routes are active. This is commonly used to create aggregate routes in OSPF, BGP, or other protocols.

Configuration Hierarchy:

The configuration for generated routes is placed under [edit routing-options], where other static and routing policies are also defined.

Command Example:

```
set routing-options generate route 10.10.0.0/16
```

Juniper Reference:

Routing Options: Juniper routers use the routing-options hierarchy to configure generated routes and other static routing behaviors.

QUESTION NO: 48

Which statement is correct about the BGP AS path when advertising routes?

- A. The order of the AS path is not significant.
- B. The local AS number is added to the end of the AS path.
- C. The order of the AS path is only significant in IBGP.
- D. The local AS number is added to the beginning of the AS path.

Answer: D

Explanation:

The BGP AS (Autonomous System) path attribute is crucial in path selection and loop prevention. Each BGP router appends its local AS number to the beginning of the AS path when it advertises a route to an external BGP (eBGP) peer.

Step-by-Step Breakdown:**AS Path Attribute:**

The AS path is a sequence of AS numbers that a route has traversed to reach a destination. Each AS adds its number to the front of the path, allowing BGP to track the route's history.

Why the Local AS is Added at the Beginning:

When advertising a route to an eBGP neighbor, a BGP router adds its own AS number to the beginning of the AS path. This ensures that the AS path reflects the route's journey accurately from the origin to the destination, and prevents loops in BGP. If the route returns to the same AS, the router will detect its AS number in the path and reject the route, preventing routing loops.

Order of the AS Path:

The order is significant because BGP uses it to select the best path. A shorter AS path is preferred, as it indicates fewer hops between the source and destination.

Juniper Reference:

AS Path Attribute: Junos devices append the local AS at the start of the AS path before advertising the route to an external peer.

QUESTION NO: 49

You want to minimize topology disruptions in your network when the rpd process restarts on a device. Which service would accomplish this task?

- A. Bidirectional Forwarding Detection (BFD)
- B. link aggregation groups
- C. graceful restart (GR)
- D. Virtual Chassis

Answer: C

Explanation:

Graceful Restart (GR) is a feature that allows a router to maintain forwarding even when the routing process (e.g., the rpd process in Junos) is restarting, minimizing disruption to the network.

Step-by-Step Breakdown:**Graceful Restart Function:**

During a GR event, the forwarding plane continues to forward packets based on existing routes, while the control plane (rpd process) is restarting. This prevents traffic loss and maintains routing stability.

Minimizing Disruptions:

GR is particularly useful in ensuring continuous packet forwarding during software upgrades or routing protocol process restarts.

Juniper Reference:

Graceful Restart in Junos: GR ensures high availability by maintaining forwarding continuity during control plane restarts, enhancing network reliability.

QUESTION NO: 50

Layer 2 interfaces operate in which two modes? (Choose two.)

- A. Access
- B. Modular

C. Trunk

D. Tagged

Answer: A,C

Explanation:

Comprehensive Detailed Step by Step Explanation with all Juniper Data Center References

Layer 2 interfaces on a switch operate in two key modes: Access and Trunk.

Step-by-Step Breakdown:

Access Mode:

Access ports are used to connect end devices, like PCs or servers, and they are assigned to a single VLAN. These interfaces handle untagged traffic and do not pass VLAN tags.

Example: A port assigned to VLAN 10 will only handle traffic for that VLAN.

Trunk Mode:

Trunk ports are used to connect switches or other networking devices that need to handle traffic from multiple VLANs. Trunk interfaces carry tagged traffic, allowing multiple VLANs to traverse the same physical link.

Trunk ports typically use 802.1Q VLAN tagging to differentiate between VLANs.

Juniper Reference:

Access and Trunk Ports: Juniper switches use these modes to manage VLAN traffic at Layer 2, with access ports handling untagged traffic and trunk ports handling tagged traffic from multiple VLANs.

QUESTION NO: 51

You are configuring an aggregate route. In this scenario, which two statements are correct? (Choose two.)

A. Reject will silently drop the traffic.

B. Discard will silently drop the traffic.

C. Reject will send an ICMP Destination Unreachable message back to the sender.

D. Discard will send an ICMP Destination Unreachable message back to the sender.

Answer: B,C

Explanation:

When configuring an aggregate route, you have options for how to handle traffic that matches the route but does not match any more specific route in the routing table. Two actions can be taken: discard and reject.

Step-by-Step Breakdown:

Discard:

The discard option will silently drop packets that match the aggregate route. No notification is sent to the sender, and the packet is simply dropped.

Reject:

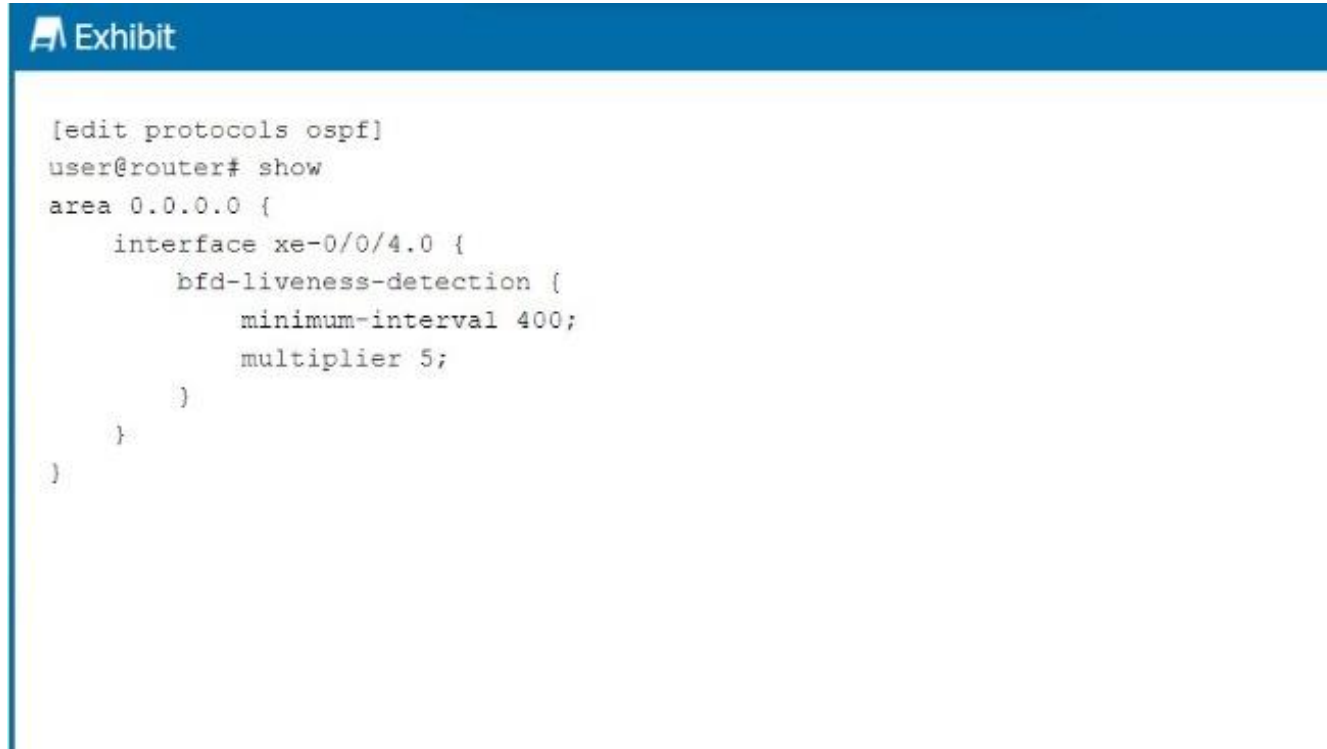
The reject option will drop the packet and also send an ICMP Destination Unreachable message back to the sender. This informs the sender that the packet could not be delivered because there is no specific route available.

Juniper Reference:

Aggregate Routes: The reject and discard next-hop options provide different levels of feedback when packets cannot be routed, and they can be used to control how unreachable destinations are handled.

QUESTION NO: 52

Exhibit:



```

[edit protocols ospf]
user@router# show
area 0.0.0.0 {
    interface xe-0/0/4.0 {
        bfd-liveness-detection {
            minimum-interval 400;
            multiplier 5;
        }
    }
}

```

Referring to the exhibit, at which interval will the interface be considered down if no hello packets are received?

- A. 2000 seconds
- B. 400 milliseconds
- C. 400 seconds
- D. 2000 milliseconds

Answer: D

Explanation:

The exhibit shows the configuration of Bidirectional Forwarding Detection (BFD) for OSPF on interface xe-0/0/4.0, with the following parameters:

minimum-interval: 400 milliseconds

multiplier: 5

Step-by-Step Breakdown:

BFD Liveness Detection:

BFD is used to detect link failures at sub-second intervals, providing faster convergence times for routing protocols like OSPF. The minimum-interval is the time between BFD control packets (in milliseconds), and the multiplier indicates how many missed BFD packets trigger a failure.

Calculating Failure Detection Time:

The failure detection interval is calculated as:

Failure Interval = minimum-interval × multiplier

In this case:

400 milliseconds × 5 = 2000 milliseconds (2 seconds)

Conclusion:

If no BFD control packets are received within 2000 milliseconds (2 seconds), the interface will be considered down, triggering OSPF to recalculate routes.

Juniper Reference:

BFD Configuration: BFD parameters such as minimum-interval and multiplier are used to fine-tune the failure detection time for faster convergence.

QUESTION NO: 53

MACsec provides protection against which two types of threats? (Choose two.)

- A. Data decryption
- B. Playback attacks
- C. Hashing attacks
- D. Man-in-the-middle attack

Answer: B,D

Explanation:

MACsec (Media Access Control Security) provides data confidentiality, integrity, and origin authenticity at Layer 2, protecting against several types of threats.

Step-by-Step Breakdown:

Man-in-the-Middle Attack Protection:

MACsec encrypts traffic at Layer 2, preventing man-in-the-middle attacks where an attacker intercepts and manipulates traffic between two communicating devices. Since the data is encrypted, any intercepted packets are unreadable.

Protection Against Playback Attacks:

MACsec also protects against playback attacks by using sequence numbers and timestamps to ensure that old, replayed packets are not accepted by the receiver.

Juniper Reference:

MACsec Configuration: Juniper devices support MACsec for securing Layer 2 communications, ensuring protection against replay and man-in-the-middle attacks in sensitive environments.

QUESTION NO: 54

Which two statements describe an IP fabric? (Choose two.)

- A. An IP fabric allows devices to always be one hop away.
- B. An IP fabric depends on Layer 2 switching.
- C. An IP fabric uses spine and leaf devices.
- D. An IP fabric provides traffic load sharing.

Answer: C,D

Explanation:

An IP fabric is a network topology designed to provide a scalable, low-latency architecture that is typically implemented in modern data centers. It uses spine and leaf switches and enables efficient traffic load sharing across the network.

Step-by-Step Breakdown:

Spine-Leaf Architecture:

Leaf Devices: These switches connect to servers and edge devices within the data center. Each leaf switch connects to every spine switch.

Spine Devices: These high-performance switches interconnect all the leaf switches. There

are no direct connections between leaf switches or spine switches. This architecture ensures that any two endpoints within the fabric are only one hop away from each other, minimizing latency.

Traffic Load Sharing:

An IP fabric leverages Equal-Cost Multipath (ECMP) to distribute traffic evenly across all available paths between leaf and spine switches, providing effective load balancing. This ensures that no single link becomes a bottleneck and that traffic is distributed efficiently across the network.

Juniper Reference:

Juniper provides QFX Series switches optimized for IP fabric topologies, allowing for scalable deployments in modern data centers.

EVPN-VXLAN: Often used in IP fabrics to extend Layer 2 services across the fabric with Layer 3 underlay, enabling both efficient routing and bridging.

QUESTION NO: 55

Which statement is correct about member interfaces when creating a LAG?

- A. The interface's MTU settings must match on all member interfaces.
- B. The interface's duplex settings and link speed must be the same on all member interfaces.
- C. Member interfaces must all be allocated on the same chassis when using a Virtual Chassis.
- D. Member interfaces must all be allocated on the same PFE.

Answer: B

Explanation:

When creating a LAG (Link Aggregation Group) in Junos, the duplex settings and link speed must be the same across all member interfaces.

Step-by-Step Breakdown:

LAG Overview:

A LAG combines multiple physical interfaces into a single logical interface to increase bandwidth and provide redundancy. All member links must act as a single cohesive unit.

Interface Requirements:

Duplex: All member interfaces must operate in the same duplex mode (either full-duplex or half-duplex). Mismatched duplex settings can cause performance issues, packet drops, or interface errors.

Link Speed: All interfaces in the LAG must have the same link speed (e.g., all interfaces must be 1 Gbps or 10 Gbps). Mismatched speeds would prevent the interfaces from functioning correctly within the LAG.

Configuration and Validation: Ensure that all member interfaces have identical settings before adding them to the LAG. These settings can be checked using the `show interfaces` command, and the LAG can be configured using:

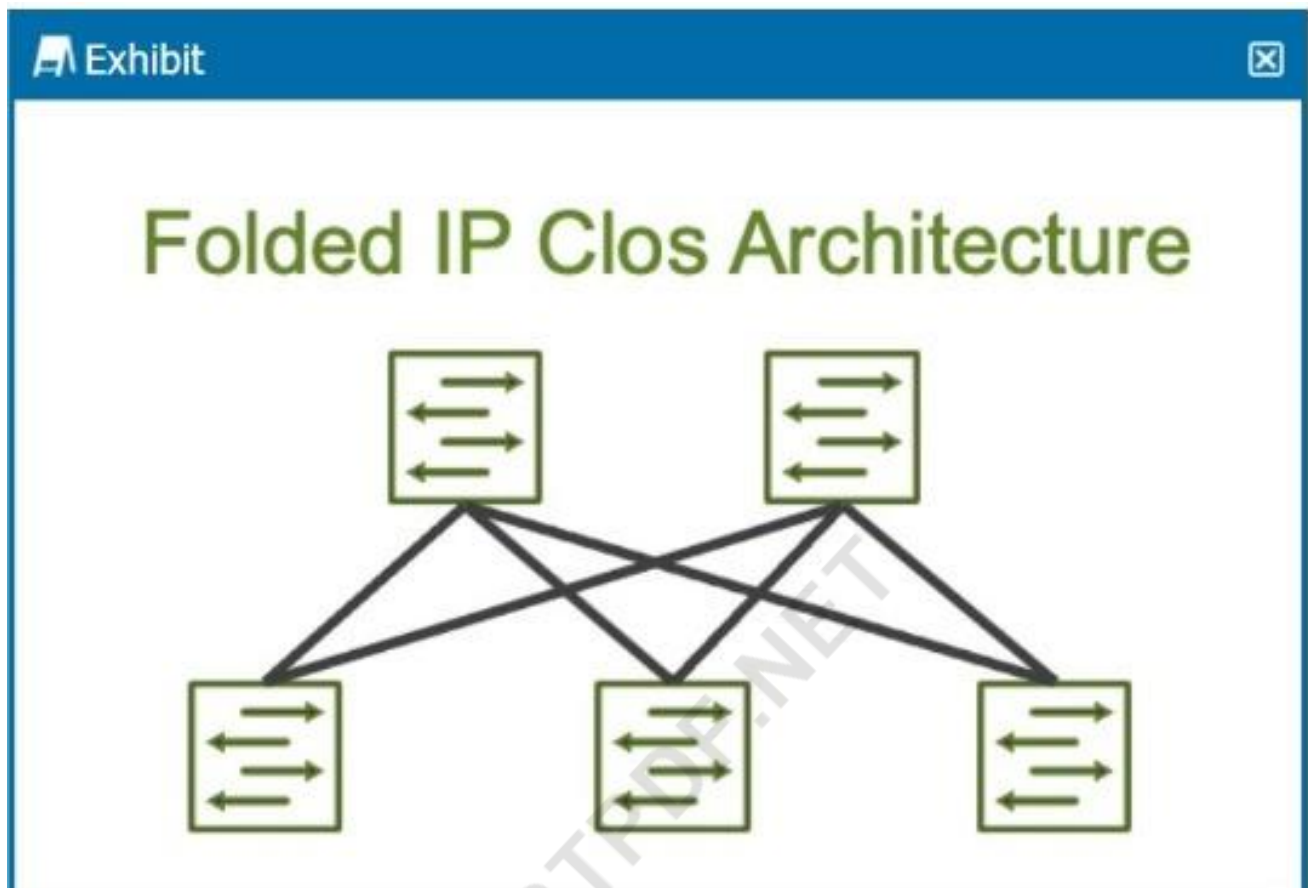
```
set interfaces ae0 aggregated-ether-options link-speed 10g
set interfaces ge-0/0/1 ether-options 802.3ad ae0
```

Juniper Reference:

LAG Configuration: Duplex and link speed must be consistent across member interfaces to ensure proper LAG operation in Juniper devices.

QUESTION NO: 56

Exhibit:



How many stages are shown in the exhibit?

- A. 2
- B. 5
- C. 6
- D. 3

Answer: D

Explanation:

The exhibit shows a Folded IP Clos Architecture, which is also referred to as a 3-stage Clos network design. This architecture typically consists of two layers of switches:

Spine Layer: The top row of switches.

Leaf Layer: The bottom row of switches.

Step-by-Step Breakdown:

Clos Architecture:

A 3-stage Clos network has two types of devices: spine and leaf. In this design, each leaf switch connects to every spine switch, providing a high level of redundancy and load balancing.

Stage Explanation:

Stage 1: The first set of leaf switches.

Stage 2: The spine switches.

Stage 3: The second set of leaf switches.

The Folded Clos architecture shown here effectively "folds" the 3-stage design by combining

the ingress and egress leaf layers into one, reducing it to two visible layers, but still maintaining the overall 3-stage architecture.

Juniper Reference:

IP Clos Architecture: The 3-stage Clos design is commonly used in modern data centers for high availability, redundancy, and scalability.

QUESTION NO: 57

Which two statements are true about how switches handle Layer 2 traffic? (Choose two.)

- A. The MAC address is learned based on the destination MAC address.
- B. The MAC address is learned based on the source MAC address.
- C. Traffic is forwarded based on the source MAC address.
- D. Traffic is forwarded based on the destination MAC address.

Answer: B,D

Explanation:

In Layer 2 switching, switches learn MAC addresses based on the source MAC address of incoming frames and forward frames based on the destination MAC address.

Step-by-Step Breakdown:

MAC Learning:

When a switch receives a frame, it records the source MAC address and the port on which it arrived. This allows the switch to know where to send traffic destined for that MAC address.

Forwarding Based on Destination:

The switch then looks at the destination MAC address and forwards the frame out of the port associated with that MAC address. If the MAC is unknown, the switch floods the frame to all ports.

Juniper Reference:

Layer 2 Switching: Juniper switches use source MAC addresses to build MAC tables and forward traffic based on the destination MAC address.

QUESTION NO: 58

Which route is preferred by the Junos OS software routing tables?

- A. Static
- B. Aggregate
- C. Direct
- D. BGP

Answer: C

Explanation:

In Junos OS, direct routes are the most preferred routes in the routing table, having the highest priority.

Step-by-Step Breakdown:

Direct Routes:

Direct routes represent networks that are directly connected to the router's interfaces. Since these routes are directly accessible, they are assigned the highest priority and always take precedence over other types of routes.

Preference Values:

Direct routes have a preference of 0, which is the most preferred in Junos. Static routes,

OSPF routes, and BGP routes have higher preference values and will only be used if there are no direct routes to the destination.

Juniper Reference:

Direct Route Preference: In Junos, direct routes are always preferred over other routes, ensuring that the router forwards traffic through locally connected networks.

QUESTION NO: 59

Which statement is correct about aggregate routes?

- A. The default next hop is discard.
- B. The default next hop is readvertise.
- C. The default next hop is resolve.
- D. The default next hop is reject.

Answer: D

Explanation:

An aggregate route is a summarized route that is created by combining multiple specific routes into a single, broader route. In Junos OS, when an aggregate route is configured, its default next hop is set to reject.

Step-by-Step Explanation:

Aggregate Route:

Aggregate routes are used to reduce the size of routing tables by representing a collection of more specific routes with a single summary route. They help improve routing efficiency and scalability, especially in large networks.

Default Next Hop Behavior:

When you configure an aggregate route in Junos OS, it has a reject next hop by default.

The reject next hop means that if a packet matches the aggregate route but there is no more specific route in the routing table for that destination, the packet will be discarded, and an ICMP "destination unreachable" message is sent to the source.

This behavior helps to prevent routing loops and ensures that traffic isn't forwarded to destinations for which there is no valid route.

Modifying Next Hop:

If needed, the next hop behavior of an aggregate route can be changed to discard (which silently drops the packet) or to another specific next hop. However, by default, the next hop is set to reject.

Juniper Reference:

Junos Command: `set routing-options aggregate route <route> reject` to configure an aggregate route with a reject next hop.

Verification: Use `show route` to verify the presence and behavior of aggregate routes.

QUESTION NO: 60

By default, which two statements are correct about BGP advertisements? (Choose two.)

- A. BGP peers advertise routes received from EBGP peers to other IBGP peers.
- B. BGP peers advertise routes received from IBGP peers to other IBGP peers.
- C. BGP peers advertise routes from EBGP peers to other IBGP peers using its own address as the next hop.
- D. BGP peers advertise routes from IBGP peers to EBGP peers using its own address as the

next hop.

Answer: A,D

Explanation:

BGP (Border Gateway Protocol) has specific rules for route advertisement between peers.

Step-by-Step Breakdown:

EBGP to IBGP Route Propagation:

BGP peers advertise routes learned from EBGP peers to IBGP peers within the same AS.

This ensures that routes learned from external networks are propagated internally within the AS.

IBGP to EBGP Route Propagation:

Routes learned from IBGP peers can be advertised to EBGP peers, but when advertising these routes, the router uses its own IP address as the next hop.

IBGP Split Horizon:

By default, IBGP peers do not advertise routes learned from one IBGP peer to another IBGP peer. This rule (IBGP split horizon) prevents routing loops within an AS.

Juniper Reference:

BGP Advertisement Rules: Junos adheres to BGP standards, where IBGP peers do not propagate routes to other IBGP peers, but EBGP peers receive IBGP routes with the advertising router as the next hop.

QUESTION NO: 61

Which statement is correct about IBGP?

- A. It requires a physical full mesh.
- B. It requires a logical full mesh.
- C. It ensures that the local and remote peers use different AS numbers.
- D. It ensures that duplicate AS numbers are not present in the AS path.

Answer: B

Explanation:

In IBGP (Internal Border Gateway Protocol), all routers within the same AS (Autonomous System) must have a logical full-mesh topology. This means that every IBGP router must be able to communicate with every other IBGP router directly or indirectly to ensure proper route propagation.

Step-by-Step Breakdown:

Logical Full Mesh:

In an IBGP setup, routers do not re-advertise routes learned from one IBGP peer to another IBGP peer. This rule is in place to prevent routing loops within the AS.

To ensure full route propagation, a logical full mesh is required, meaning every IBGP router must peer with every other IBGP router in the AS. This can be done either directly or via route reflection or confederation.

Physical Full Mesh Not Required:

The physical topology does not need to be a full mesh, but the BGP peering relationships must form a logical full mesh. Techniques like route reflectors or BGP confederations can reduce the need for manual full-mesh peering.

Juniper Reference:

IBGP Configuration: IBGP logical full mesh requirements can be simplified using route

reflectors to avoid the complexity of manually configuring many IBGP peers.

QUESTION NO: 62

When evaluating BGP routes, what will be evaluated first?

- A. The local preference value
- B. The AS path
- C. The MED value
- D. The origin value

Answer: A

Explanation:

In BGP (Border Gateway Protocol), when evaluating multiple routes to the same destination, the first attribute that is considered is the local preference value. The local preference is a BGP attribute used to influence outbound routing decisions within an Autonomous System (AS).

Step-by-Step Breakdown:

Local Preference:

The local preference attribute is used to determine which path is preferred for traffic leaving the AS. The higher the local preference value, the more preferred the route.

BGP Path Selection:

The BGP path selection process evaluates the following attributes in this order:

Local Preference (higher is preferred)

AS Path (shorter is preferred)

Origin (IGP > EGP > incomplete)

MED (Multi-Exit Discriminator) (lower is preferred)

Juniper Reference:

BGP Path Selection: In Junos, the local preference attribute is the first to be evaluated when determining the best path for outbound traffic.

QUESTION NO: 63

Which operation mode command will display the mapping between the VLAN ID and ports on a switch?

- A. show route
- B. show ethernet-switching table
- C. show interfaces terse
- D. show vlans

Answer: D

Explanation:

To display the mapping between VLAN IDs and ports on a Juniper switch, the show vlans command is used.

Step-by-Step Breakdown:

VLAN Information:

The show vlans command displays detailed information about VLAN configurations, including the VLAN ID, associated interfaces (ports), and VLAN membership.

Command Example:

show vlans

This command will provide an output listing each VLAN, its ID, and the interfaces associated with the VLAN, enabling network engineers to quickly verify VLAN to port mappings.

Juniper Reference:

VLAN Verification: Use the show vlans command to verify which VLANs are configured on the switch and the ports that are members of those VLANs.

QUESTION NO: 64

When a MAC limiting violation occurs, the switch performs which two actions by default? (Choose two.)

- A. No logging takes place.
- B. It causes Layer 2 loops.
- C. The port is disabled.
- D. It drops the packet.

Answer: C,D

Explanation:

When a MAC limiting violation occurs on a Juniper switch, the switch will perform the following actions by default:

Step-by-Step Breakdown:

Port Disabled:

When the number of MAC addresses on an interface exceeds the configured limit, the port is automatically disabled to prevent further violations. This is a protective mechanism to prevent MAC address flooding.

Packet Dropped:

Additionally, packets from the violating MAC address are dropped to prevent any further communication from that address. This ensures that only valid MAC addresses are allowed to communicate through the interface.

Example Configuration:

```
set ethernet-switching-options secure-access-port interface <interface-name> mac-limit 5
```

If more than five MAC addresses are learned, the port is disabled, and excess packets are dropped.

Juniper Reference:

MAC Limiting: When the switch detects a MAC limiting violation, it disables the port and drops further packets from the violating MAC addresses to maintain network security.

QUESTION NO: 65

Which statement is correct about per-flow load balancing?

- A. Packets associated with the same flow are sent through different egress ports.
- B. The packets are guaranteed to arrive at their destination in a different order in which they were sent.
- C. Packets associated with the same flow are sent through the same egress port.
- D. The packets are guaranteed to arrive at their destination in the same order in which they were sent.

Answer: C

Explanation:

Per-flow load balancing ensures that packets within the same flow are always forwarded over

the same path, ensuring that packet order is preserved.

Step-by-Step Breakdown:

Flow Definition:

A flow is typically defined by a combination of packet attributes like source/destination IP, source/destination port, and protocol type. Packets that belong to the same flow are routed over the same path to avoid reordering.

Per-Flow Behavior:

In per-flow load balancing, the hashing algorithm ensures that all packets in a particular flow use the same egress port, maintaining order across the network.

Juniper Reference:

Load Balancing in Juniper: This method ensures that flows are balanced across multiple paths while preventing packet reordering within a single flow.

QUESTION NO: 66

Which of the following are true regarding port modes in a VLAN setup? (Choose two)

- A. An access port can carry traffic for multiple VLANs.
- B. A trunk port can carry traffic for multiple VLANs.
- C. An access port carries traffic only for a single VLAN.
- D. A trunk port is used exclusively for inter-switch links.

Answer: B C

QUESTION NO: 67

What statement is true about load-balancing?

- A. Load balancing is only supported after applying an advanced license.
- B. Load balancing on Juniper devices is processed per-packet.
- C. Load balancing on Juniper devices is processed per-flow.
- D. Load balancing is only supported by dynamic routing protocols.

Answer: C

QUESTION NO: 68

VLANs help in reducing which of the following? (Choose two)

- A. Collision domains
- B. Broadcast domains
- C. Routing complexity
- D. Encryption requirements

Answer: A B

QUESTION NO: 69

What is an aggregate route?

- A. a static route with only one contributing route
- B. a group of IP addresses to which a router will not forward traffic
- C. a mechanism that allows multiple routes to be summarized into a single route
- D. often called the route of last resort

Answer: C

QUESTION NO: 70

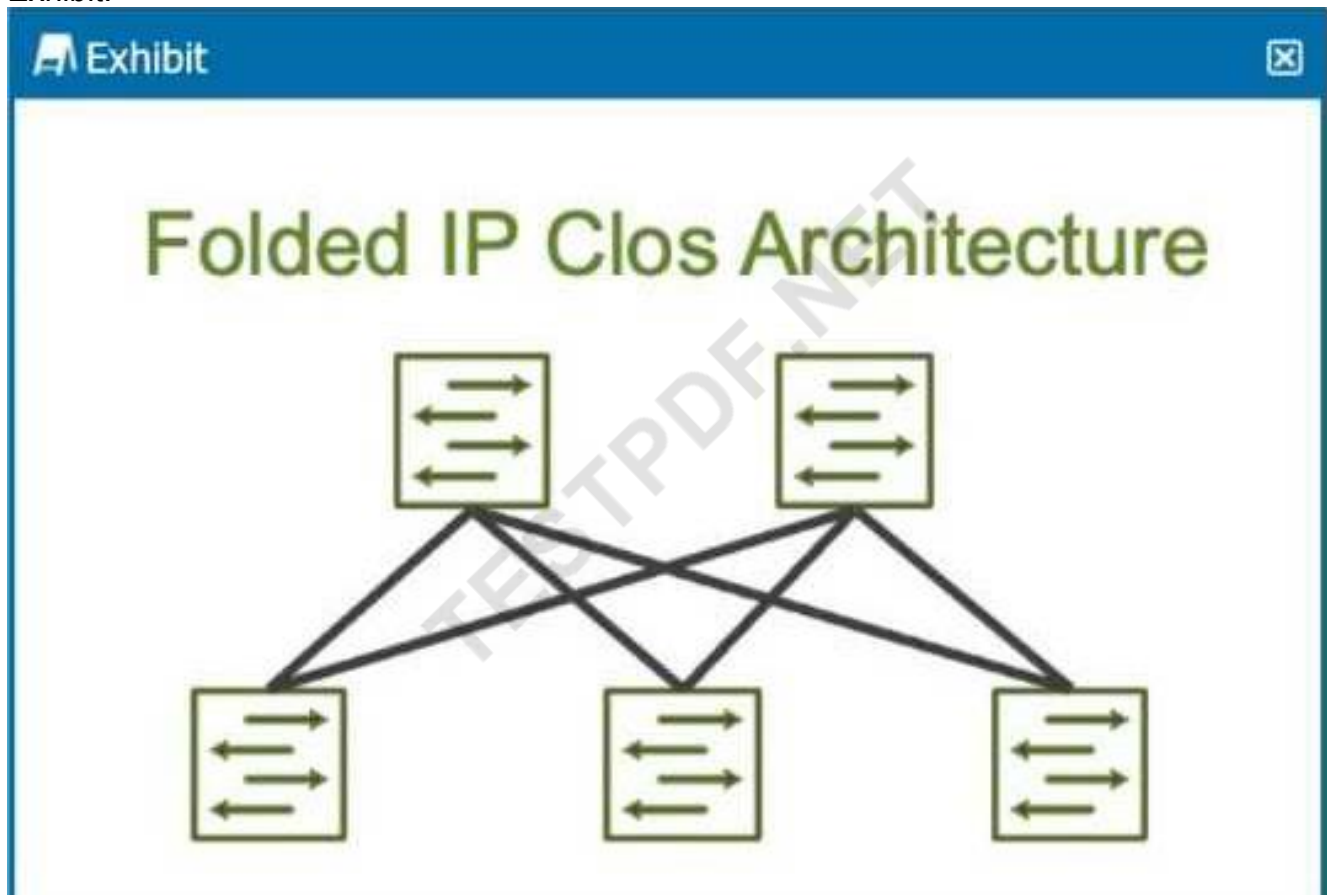
In the context of Link Aggregation Groups (LAG), what is the primary purpose of LACP?

- A. To provide a method for encrypting data packets
- B. To dynamically manage the bundling of several physical ports
- C. To reduce the speed of data transfer
- D. To increase the error rate in data transmission

Answer: B

QUESTION NO: 71

Exhibit:



How many stages are shown in the exhibit?

- A. 2
- B. 5
- C. 6
- D. 3

Answer: D

Explanation:

The exhibit shows a Folded IP Clos Architecture, which is also referred to as a 3-stage Clos network design. This architecture typically consists of two layers of switches: Spine Layer:

The top row of switches.

Leaf Layer: The bottom row of switches.

Step-by-Step Breakdown:

Clos Architecture:

A 3-stage Clos network has two types of devices: spine and leaf. In this design, each leaf switch connects to every spine switch, providing a high level of redundancy and load balancing. Stage Explanation:

Stage 1: The first set of leaf switches.

Stage 2: The spine switches.

Stage 3: The second set of leaf switches.

The Folded Clos architecture shown here effectively "folds" the 3-stage design by combining the ingress and egress leaf layers into one, reducing it to two visible layers, but still maintaining the overall 3-stage architecture.

Juniper

Reference: IP Clos Architecture: The 3-stage Clos design is commonly used in modern data centers for high availability, redundancy, and scalability.

QUESTION NO: 72

In troubleshooting a Virtual Chassis setup, what is a common issue to look for?

- A. Incompatible software versions between member switches
- B. Too many switches in a single Virtual Chassis configuration
- C. Use of copper cables for inter-switch connections
- D. Activation of all ports for Virtual Chassis connections

Answer: A

QUESTION NO: 73

What is the first route selection mechanism for IBGP?

- A. as path
- B. MED
- C. local preference
- D. origin

Answer: C

QUESTION NO: 74

When configuring BGP, which command is commonly used to define a BGP neighbor?

- A. bgp set neighbor
- B. neighbor <IP_ADDRESS> remote-as <AS_NUMBER>
- C. set bgp peer
- D. add bgp connection

Answer: B

QUESTION NO: 75

What are the different OSPF area types? (Choose two)

- A. Stub Area
- B. Not-So-Stubby Area (NSSA)
- C. Default-Free Zone
- D. Virtual Link Area

Answer: A B

QUESTION NO: 76

During a BGP Graceful Restart, what happens to the routes learned from the restarting neighbor?

- A. They are immediately removed from the routing table.
- B. They are marked as stale but kept in the routing table.
- C. They are advertised to other BGP neighbors.
- D. Their metric is increased to prevent their use.

Answer: B

QUESTION NO: 77

Bidirectional Forwarding Detection (BFD) is primarily used for:

- A. Encrypting data packets
- B. Detecting slow links in the network
- C. Quick failure detection in forwarding paths
- D. Aggregating multiple links

Answer: C

QUESTION NO: 78

What roles do the Designated Router (DR) and Backup Designated Router (BDR) play in OSPF?

- A. They redistribute routes between OSPF and non-OSPF networks.
- B. They are responsible for generating LSAs for external routes.
- C. They reduce the number of adjacencies on broadcast networks.
- D. They determine the shortest path first algorithm.

Answer: C

QUESTION NO: 79

What is a VLAN?

- A. A local network using only virtual machines
- B. A broadcast domain, identified by a VLAN ID
- C. A multicast domain, identified by a VLAN ID
- D. A broadcast domain, using only virtual machines

Answer: B

QUESTION NO: 80

Which BGP message type is used to establish peering sessions?

- A. Update
- B. Notification
- C. Open
- D. Keepalive

Answer: C

QUESTION NO: 81

What happens if a frame with a known destination MAC address enters a switch?

- A. The egress PFE performs a MAC address lookup and determines the ingress PFE and port.
- B. The ingress PFE sends the header information to the RE.
- C. The ingress PFE performs a MAC address lookup and determines the egress PFE and port.
- D. The MAC address is added to the bridge table.

Answer: C

QUESTION NO: 82

What is the main advantage of using an Overlay Network in a data center?

- A. It simplifies physical network design by allowing multiple virtual topologies over it.
- B. It replaces the need for physical network infrastructure.
- C. It offers higher physical bandwidth capacity.
- D. It provides direct Internet access to applications.

Answer: A

QUESTION NO: 83

Which command on a router is used to view the current routing table?

- A. show routing-table
- B. show route
- C. display ip routing-table
- D. check route information

Answer: B

QUESTION NO: 84

In the context of data center networking, what is the primary purpose of an Underlay Network?

- A. To provide a physical infrastructure for an Overlay Network.
- B. To handle external network connections exclusively.
- C. To facilitate virtual machine migrations.
- D. To encrypt data traffic between servers.

Answer: A

QUESTION NO: 85

Which two prefixes are contributing routes to a 10.20.0.0/18 aggregate route? (Choose two)

- A. 10.20.0.0/24
- B. 10.20.60.0/23
- C. 10.20.0.0/16
- D. 10.20.100.0/24

Answer: A B

QUESTION NO: 86

Which LSA types are associated with OSPF? (Choose three)

- A. Router LSA (Type 1)
- B. Network LSA (Type 2)
- C. Summary LSA (Type 3)
- D. External LSA (Type 5)

Answer: A B C

QUESTION NO: 87

Which two statements are correct about aggregated Ethernet bundles? (Choose two.)

- A. You must enable the chassis to support aggregated Ethernet interfaces.
- B. You must purchase a services license for the chassis to support aggregate Ethernet interfaces.
- C. You must specify which interfaces are members of a specific aggregated Ethernet bundle.
- D. You must enable LACP when configuring aggregated Ethernet bundles.

Answer: A C

QUESTION NO: 88

What is an advantage of using BFD?

- A. BFD is so lightweight that each protocol will generate its own unique BFD session by default, to replace its own hello messages.
- B. BFD built to handle the routing table of the entire Internet, and can safely redistribute it into OSPF.
- C. One BFD monitoring session can be used by multiple protocols.
- D. BFD is built to handle the routing table of the entire Internet, unlike IGPS.

Answer: C

QUESTION NO: 89

When configuring High Availability components, which of the following practices are recommended? (Choose two)

- A. Regularly update firmware to non-recommended versions.
- B. Ensure physical redundancy of critical components.
- C. Use a single path for data and control traffic.
- D. Implement regular testing of failover mechanisms.

Answer: B D

QUESTION NO: 90

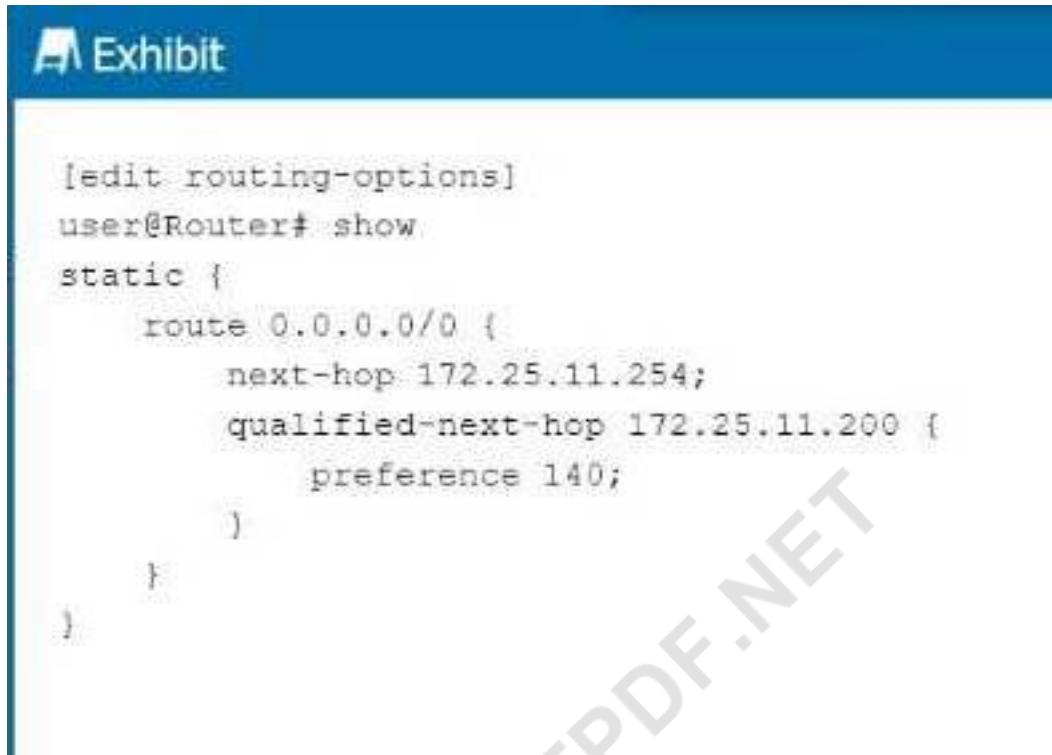
Which Junos operational mode command is used to troubleshoot issues with VLANs?

- A. show vlans
- B. monitor traffic
- C. show ethernet-switching table
- D. show interfaces

Answer: C

QUESTION NO: 91

Exhibit:

The exhibit shows a Junos configuration snippet for static routes. It starts with '[edit routing-options]' followed by 'user@Router# show'. Under the 'static' block, there is a 'route 0.0.0.0/0' block. Inside this block, there are two 'next-hop' entries: 'next-hop 172.25.11.254;' and 'qualified-next-hop 172.25.11.200 { preference 140; }'. The configuration is enclosed in curly braces to indicate it's a snippet from a larger file.

```
[edit routing-options]
user@Router# show
static {
    route 0.0.0.0/0 {
        next-hop 172.25.11.254;
        qualified-next-hop 172.25.11.200 {
            preference 140;
        }
    }
}
```

Referring to the exhibit, what is the route preference of the 172.25.11.254 next hop?

- A. 5
- B. 10
- C. 130
- D. 140

Answer: A

Explanation:

In the exhibit, we see two next-hop addresses for the default static route (0.0.0.0/0):

The first next hop is 172.25.11.254, with no specified preference.

The second next hop is 172.25.11.200, with a specified preference of 140.

Step-by-Step Breakdown:

Default Static Route Preference:

If no preference is explicitly set for a next hop in Junos, it defaults to 5 for static routes.

Determining Preference:

In this case, the next hop 172.25.11.254 does not have an explicit preference defined, so it will use the default value of 5. The second next hop has a preference of 140, which is higher, meaning it will only be used if the primary next hop is unavailable. Juniper Reference: Static

Route Preference: In Junos, the default preference for static routes is 5, and this value is applied unless overridden by the preference parameter.

QUESTION NO: 92

What is Integrated Routing and Bridging (IRB) primarily used for in a Junos device?

- A. To separate VLAN traffic.
- B. To provide inter-VLAN routing.

- C. For aggregating multiple interfaces.
- D. For implementing security policies.

Answer: B

QUESTION NO: 93

Which of the following are correct about OSPF router ID selection? (Choose two)

- A. It can be manually configured.
- B. It is chosen based on the highest IP address of any active interface by default.
- C. It is always the IP address of the loopback interface.
- D. It is selected based on the lowest MAC address.

Answer: A B

QUESTION NO: 94

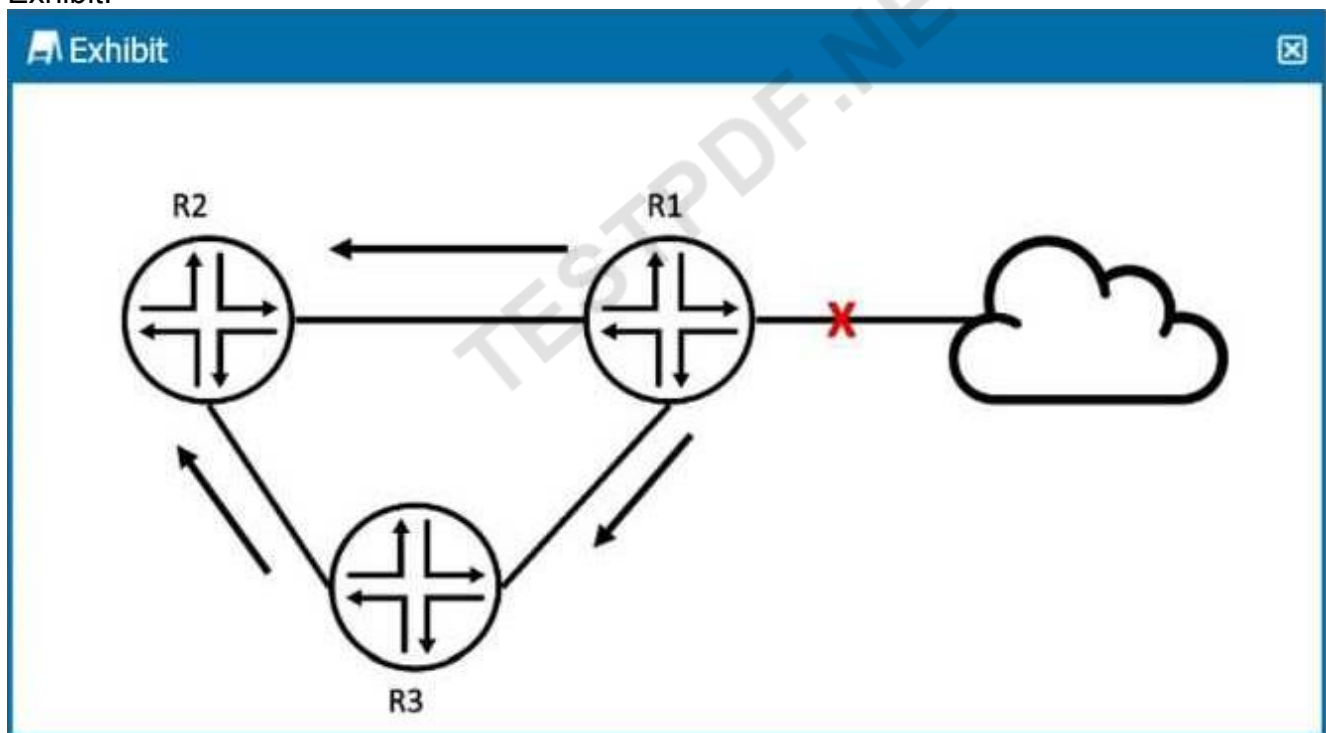
Which statement is correct about an IP fabric?

- A. Transit traffic is passed through the network based on the original frame's destination MAC address.
- B. Load sharing is provided by the routing protocol used.
- C. Loop prevention is provided by the underlying spanning tree protocol used.
- D. The entire IP fabric is one large broadcast domain.

Answer: B

QUESTION NO: 95

Exhibit:



R2 received an OSPF update from R1, and it received the same update from R3. Referring to the exhibit, what will R2 do?

- A. R2 ignores the update from R1.
- B. R2 does nothing with R3's update.

- C. R2 ignores the update from R3.
- D. R2 acknowledges R3 and discards it.

Answer: C

Explanation:

In the exhibit, R2 receives the same OSPF update from both R1 and R3. OSPF has mechanisms to prevent unnecessary processing of duplicate LSAs (Link-State Advertisements).

Step-by-Step Breakdown:

OSPF LSA Processing:

OSPF uses LSAs to exchange link-state information between routers. When a router receives an LSA, it checks if it already has a copy of the LSA in its Link-State Database (LSDB).

Duplicate LSAs:

If R2 has already received and processed the update from R1, it will ignore the update from R3 because it already has the same LSA in its database. OSPF uses the concept of flooding, but it does not reprocess LSAs that it already knows about.

R2 Behavior:

R2 will keep the update from R1 (the first one it received) and will ignore the same LSA from R3, as it is already in the LSDB.

Juniper

Reference: OSPF LSA Processing: Junos adheres to OSPF standards, ensuring that duplicate LSAs are not processed multiple times to avoid unnecessary recalculations.

QUESTION NO: 96

What is a key characteristic of traditional multi-tier data center architectures?

- A. They primarily use a flat network design.
- B. They consist of a modular, scalable design.
- C. They are organized in a hierarchical model with distinct layers.
- D. They exclusively use Layer 3 routing.

Answer: C

QUESTION NO: 97

Which of the following is a key benefit of implementing High Availability (HA) in a network?

- A. Reduced network security
- B. Increased network downtime
- C. Enhanced data redundancy
- D. Slower data recovery

Answer: C

QUESTION NO: 98

When configuring OSPF, what is the purpose of defining areas?

- A. To specify different encryption methods.
- B. To control route advertisement and manage traffic.
- C. To assign different IP address ranges.

D. To provide network redundancy.

Answer: B

QUESTION NO: 99

How can OSPF be configured to prefer one route over another?

- A. By setting a higher cost on the preferred route.
- B. By configuring route redistribution.
- C. By adjusting interface bandwidth.
- D. By setting a lower cost on the preferred route.

Answer: D

QUESTION NO: 100

What is the primary purpose of using aggregate routes in a network?

- A. To dynamically adjust to network changes
- B. To summarize and reduce the number of routes in the routing table
- C. To filter unwanted traffic
- D. To statically define a route to a specific destination

Answer: B

QUESTION NO: 101

In networking, what is a RIB group primarily used for?

- A. To balance the load across multiple links
- B. To filter routes based on specific criteria
- C. To group multiple Routing Information Bases together for a common purpose
- D. To dynamically generate routing entries

Answer: C

QUESTION NO: 102

Which two protocols can you use for a data center underlay network? (Choose two)

- A. EBGp
- B. OSPF
- C. VXLAN
- D. EVPN

Answer: A B

QUESTION NO: 103

Which of the following BGP attributes is considered first in the route selection process?

- A. Local Preference
- B. AS Path Length
- C. Origin
- D. MED

Answer: A

QUESTION NO: 104

What is the primary function of a VLAN?

- A. To encrypt network traffic.
- B. To provide a routing mechanism between different networks.
- C. To segment a network into multiple broadcast domains.
- D. To control network traffic based on MAC addresses.

Answer: C

QUESTION NO: 105

In OSPF, which command is commonly used to monitor neighbor relationships?

- A. show ip ospf neighbors
- B. show ospf database
- C. show ip route ospf
- D. show ospf interfaces

Answer: A

QUESTION NO: 106

What is the main advantage of using MACsec?

- A. It increases the speed of the network.
- B. It provides Layer 2 encryption for data confidentiality and integrity.
- C. It creates multiple virtual networks over a single physical infrastructure.
- D. It allows for the use of longer cable runs.

Answer: B

QUESTION NO: 107

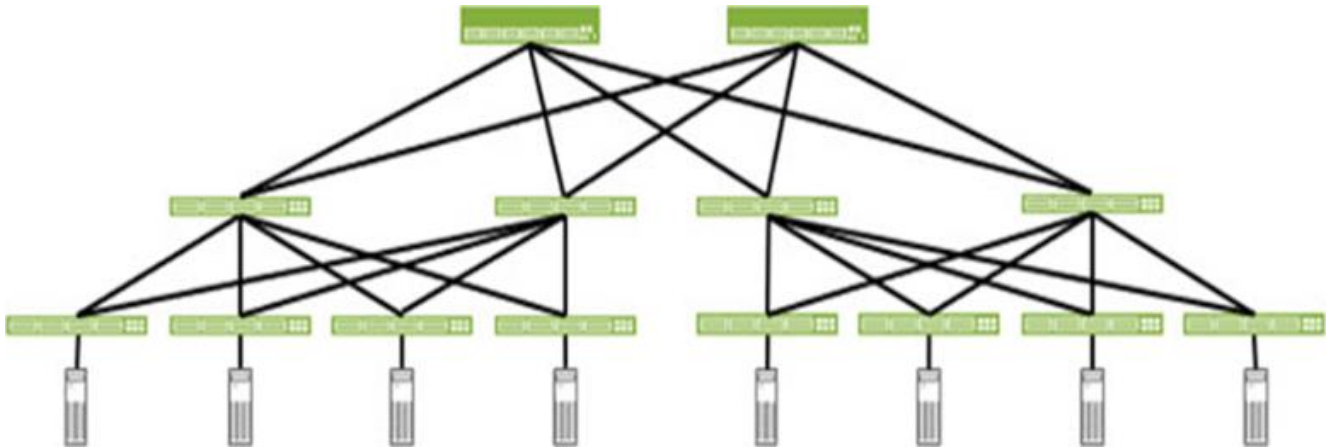
What is the primary use of Martian addresses in network routing?

- A. To define addresses used for multicasting.
- B. To identify addresses that should be excluded from routing.
- C. To configure virtual IP addresses for load balancing.
- D. To create a backup routing plan.

Answer: B

QUESTION NO: 108

Referring to the exhibit, how many stages are in this IP fabric?



- A. 4
- B. 2
- C. 5
- D. 3

Answer: C

QUESTION NO: 109

You are troubleshooting a downed BGP session.

Exhibit
✕

```

user@router> show bgp summary | match "AS|Connect|Active"
Peer          AS      InPkt   OutPkt   OutQ   Flaps Last Up/Dwn  Connect
State|#Active/Received/Accepted/Damped...
10.10.55.2    1111      0        0        0      10 1w0d 3:51:17 Connect
  
```

Referring to the exhibit, what is the cause of the problem?

- A. The UDP session between the peers has not been established.
- B. The local peer has sent an Open message but not received one from the remote peer.
- C. The TCP session between the peers has not been established.
- D. The local peer has sent an Update message but not received one from the remote peer.

Answer: C

Explanation:

The BGP session in the exhibit shows the state as Connect, which indicates that the TCP session between the BGP peers has not been fully established. Step-by-Step Breakdown: BGP State "Connect":

The Connect state is the second stage in the BGP finite state machine (FSM). At this stage, BGP is trying to establish a TCP session with the peer, but the session has not yet been successfully established.

A successful TCP three-way handshake (SYN, SYN-ACK, ACK) is required before BGP can progress to the OpenSent state, where the peers exchange BGP Open messages.

Possible Causes:

A firewall blocking TCP port 179.

Incorrect IP addresses or network connectivity issues between the BGP peers. Juniper

Reference: BGP Troubleshooting: In Junos, if a BGP session is stuck in the Connect state,

the issue is likely due to a failure in establishing the underlying TCP connection.

QUESTION NO: 110

Martian addresses are:

- A. Special routes for interplanetary communication.
- B. Addresses used in MPLS.
- C. Unroutable addresses that are generally filtered by routers.
- D. Addresses used for load balancing.

Answer: C

QUESTION NO: 111

Which VLAN tagging method inserts a 4-byte tag into the Ethernet frame?

- A. ISL
- B. Dot1Q
- C. Q-in-Q
- D. MPLS

Answer: B

QUESTION NO: 112

How does VLAN tagging benefit network segmentation?

- A. By encrypting data traffic between VLANs.
- B. By ensuring the same IP address range can be used in different VLANs.
- C. By enabling the identification and separation of traffic belonging to different VLANs.
- D. By providing a backup path for data in case of link failure.

Answer: C

QUESTION NO: 113

EVPN/VXLAN is primarily used in modern data centers to:

- A. Enhance the physical security of the data center.
- B. Replace traditional routing protocols.
- C. Increase the physical distance between data centers.
- D. Provide Layer 2 connectivity over a Layer 3 network.

Answer: D

QUESTION NO: 114

Which OSPF troubleshooting tool is used to check the OSPF process and neighborship?

- A. Traceoptions
- B. Ping
- C. Route summarization
- D. SNMP

Answer: A

QUESTION NO: 115

Exhibit:

 Exhibit

```
{master:0}[edit switch-options]
user@switch# show
interface ge-0/0/1.0 {
    persistent-learning;
}
```

Referring to the exhibit, which behavior does this configuration enable on the ge-0/0/1.0 interface?

- A. This configuration enables a MAC address learned on the interface to be persistently retained in the Ethernet-switching table, even after a reboot.
- B. This configuration enables the device to place a MAC address that persistently causes network errors into a special protected VLAN.
- C. This configuration enables the device to shut down the interface when a particular MAC address persistently sends broadcast traffic.
- D. This configuration enables the interface to learn and remember MAC addresses, until the device is rebooted.

Answer: A

Explanation:

The configuration in the exhibit shows the persistent-learning feature enabled on interface ge-0/0/1.0.

Step-by-Step Breakdown:

Persistent Learning:

Persistent-learning ensures that the MAC addresses learned on the interface are retained in the Ethernet-switching table, even after a device reboot. This prevents the need to re-learn MAC addresses after the device restarts, improving stability and reducing downtime.

Use Case:

This feature is particularly useful in environments where the re-learning of MAC addresses could cause temporary disruptions or delays in communication, such as in critical Layer 2 network segments.

Command Example:

```
set switch-options interface ge-0/0/1.0 persistent-learning
```

Juniper

Reference: Persistent MAC Learning: In Junos, enabling persistent-learning ensures that learned MAC addresses are not lost during reboots, contributing to smoother network operations in environments where stability is crucial.

QUESTION NO: 116

What does VXLAN stand for in the context of EVPN/VXLAN?

- A. Virtual Extensible VLAN
- B. Virtualized Ethernet VPN
- C. Very Extended Virtual LAN
- D. Virtual X-Series LAN

Answer: A

QUESTION NO: 117

When configuring MAC address filtering on a Junos switch, which feature is primarily used?

- A. Firewall filters
- B. RADIUS
- C. Access Control Lists (ACLs)
- D. Port security

Answer: D

QUESTION NO: 118

What is the default Junos behavior of access ports?

- A. Allow only untagged traffic
- B. Allow only tagged traffic
- C. Allow both tagged and untagged traffic
- D. Allow a maximum of 80 percent untagged traffic

Answer: A

QUESTION NO: 119

BGP is primarily used for which purpose in a network?

- A. Internal routing within a single autonomous system
- B. Connecting different autonomous systems and route sharing
- C. Providing Layer 2 connectivity
- D. Encrypting data traffic between networks

Answer: B

QUESTION NO: 120

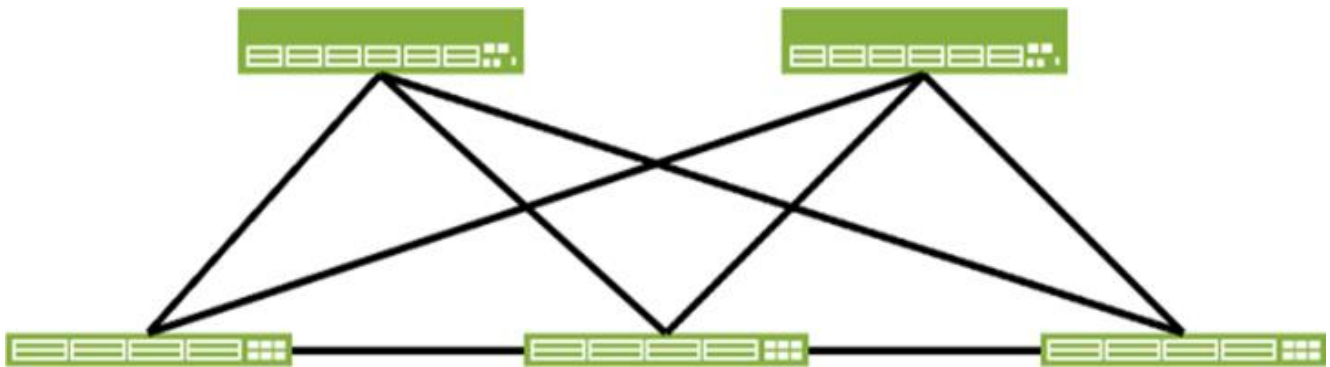
Which of the following are features of storm control on Junos switches? (Choose two)

- A. It can be configured to shut down a port during a storm.
- B. It supports thresholds for broadcast, multicast, and unknown unicast traffic.
- C. It encrypts broadcast and multicast traffic.
- D. It automatically reroutes traffic during a storm.

Answer: A B

QUESTION NO: 121

Referring to the exhibit, what needs to be done to make this a valid IP fabric?



- A. Remove the connection between the leaf nodes
- B. Remove one of the redundant links between a leaf node and a spine node
- C. Add two connections between the spine nodes
- D. Add a connection between the spine nodes

Answer: A

QUESTION NO: 122

Referring to the exhibit, which two statements are true? (Choose two.)

```
{master:0}[edit protocols bgp]
user@switch# show
group DC_overlay {
  multihop;
  local-address 192.168.100.1;
  family evpn {
    signaling;
  }
  neighbor 192.168.100.2;
}
```

- A. The configuration will not pass the commit check.
- B. The BGP group is configured to attempt an EBGp peering session with 192.168.100.2.
- C. The configuration will pass the commit check.
- D. The BGP group is configured to attempt an EBGp peering session with 192.168.100.1.

Answer: A B

QUESTION NO: 123

You have configured a load balancing policy.

Which statement is correct about applying the policy?

- A. The policy is applied as an export policy under the routing protocol's hierarchy.

- B. The policy is applied as an export policy under routing-options forwarding table.
- C. The policy is applied as an import policy under the routing protocol's hierarchy.
- D. The policy is applied as an import policy under routing-options forwarding table.

Answer: B

QUESTION NO: 124

What is the purpose of using route reflectors in BGP?

- A. To encrypt BGP messages
- B. To reduce the number of required IBGP sessions in a network
- C. To increase the speed of BGP convergence
- D. To automatically select the best route

Answer: B

QUESTION NO: 125

Which two statements describe an OSPF ABR? (Choose two.)

- A. By default, an ABR does not need to have any links in the backbone area.
- B. By default, an ABR is an OSPF router with links in two areas.
- C. By default, an ABR is an OSPF router with all of its links in one area.
- D. By default, an ABR must have one link in the OSPF backbone area.

Answer: B D

QUESTION NO: 126

Which two statements are true with regards to MACsec?

- A. It provides authentication for an Ethernet traffic between two network nodes.
- B. It does not provide encryption for an Ethernet traffic between two network nodes.
- C. It does not provide authentication for an Ethernet traffic between two network nodes.
- D. It provides encryption for an Ethernet traffic between two network nodes.

Answer: A D

QUESTION NO: 127

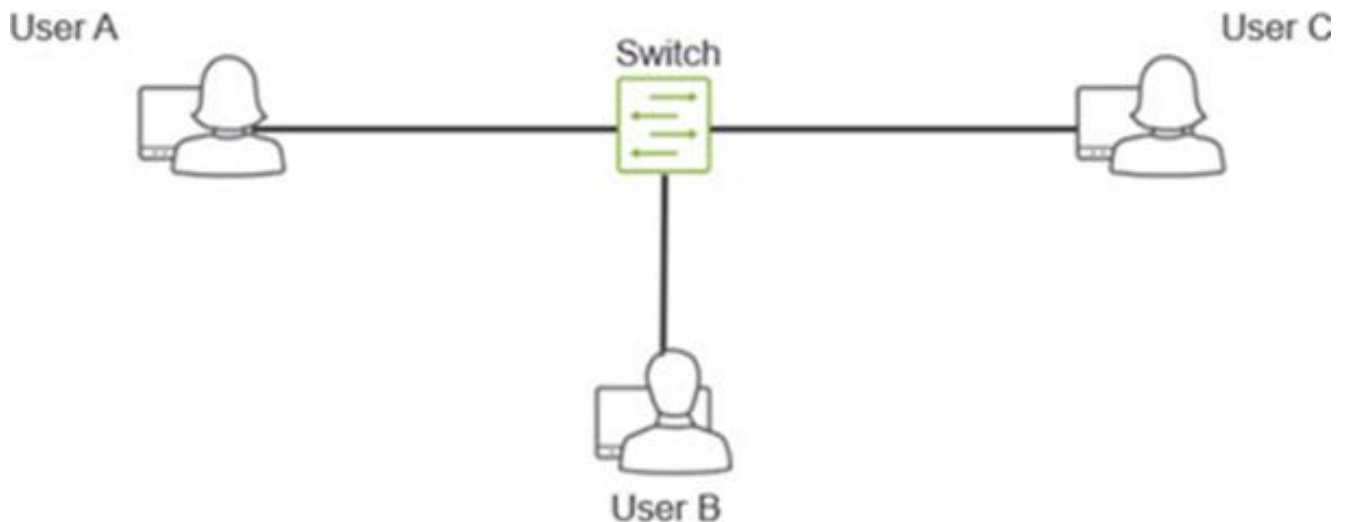
Which of the following statements are true regarding Graceful Restart (GR)? (Choose two)

- A. It is used to increase the convergence time of the network.
- B. It decreases network stability during protocol restarts.
- C. It requires support from both the restarting router and its neighbors.
- D. It helps in maintaining uninterrupted forwarding during a routing process restart.

Answer: C,D

QUESTION NO: 128

Referring to the exhibit, how many collision domains are present in the topology?



- A. 3
- B. 0
- C. 2
- D. 1

Answer: A

QUESTION NO: 129

What protocol and port does BGP use for control plane communication?

- A. protocol TCP, port 179
- B. protocol UDP, port 179
- C. protocol TCP, port 159
- D. protocol UDP, port 159

Answer: A

QUESTION NO: 130

Which two statements are correct about BGP? (Choose two.)

- A. IBGP peers do not advertise routes learned from IBGP peers to other IBGP peers.
- B. EBGP uses the AS path to avoid loops.
- C. IBGP sessions are typically established between physical router interfaces.
- D. EBGP sessions are typically established between router loopback interfaces.

Answer: A B

QUESTION NO: 131

Which statement is correct about the OSPF link-state database?

- A. Each router within the same area must maintain a unique copy of the link-state database.
- B. Each router within the same area must maintain an identical copy of the link-state database.
- C. Each router within the OSPF domain must maintain a unique copy of the link-state database.
- D. Each router within the OSPF domain must maintain an identical copy of the link-state database.

Answer: B

QUESTION NO: 132

Referring to the exhibit, which statement is correct?

```
{master:0}
```

```
user@switch> show vlans
```

Routing instance	VLAN name	Tag	Interfaces
default-switch	default	1	ge-0/0/0.0* ge-0/0/1.0
default-switch	GREEN-10	10	ge-0/0/12.0* ge-0/0/8.0*
default-switch	BLUE-20	20	ge-0/0/12.0* ge-0/0/9.0*

- A. The ge-0/0/0 interface has been explicitly assigned to a VLAN named default.
- B. The default VLAN is a user defined VLAN name.
- C. The asterisk indicates that the interface is active.
- D. The ge-0/0/12 interfaces is an access port.

Answer: C

QUESTION NO: 133

Referring to the exhibit, which statement is true?


```
{master:0}
```

```
user@switch> show bfd session extensive
```

```

                        Detect Transmit
Address      State  Interface  Time  Interval Multiplier
172.16.1.1   Up    xe-0/0/0.0  2.700  0.900    3
Client BGP, TX interval 0.900, RX interval 0.900
Session up time 00:00:39
Local diagnostic None, remote diagnostic None
Remote state Up, version 1
Session type: Single hop BFD
Min async interval 0.900, min slow interval 1.000
Adaptive async TX interval 0.900, RX interval 0.900
Local min TX interval 0.900, minimum RX interval 0.900, multiplier 3
Remote min TX interval 0.900, min RX interval 0.900, multiplier 3
Local discriminator 16, remote discriminator 16
Echo TX interval 0.000, echo detection interval 0.000
Echo mode disabled/inactive
Remote is control-plane independent
Session ID: 0x140

```

- A. BFD will bring down the BGP session if no BFD packets are received for 2700 seconds.
- B. BFD packets are being sent every second.
- C. BFD will bring down the BGP session if no BFD packets are received for 2700 milliseconds.
- D. BFD only records a syslog message if three BFD packets are missed.

Answer: C

QUESTION NO: 134

What are two characteristics of a typical EBGp underlay fabric? (Choose two.)

- A. Each device in an EBGp fabric will be configured in its own unique private AS.
- B. Each device in an EBGp fabric will be configured to be part of the same private AS.
- C. An EBGp fabric relies on an IGP to advertise loopback IPs.
- D. An EBGp fabric does not require an IGP to advertise loopback IPs.

Answer: A D

QUESTION NO: 135

In Ethernet switching, what is the primary function of a storm control policy?

- A. To prevent loops in the network.
- B. To restrict the rate of certain types of traffic.

- C. To filter MAC addresses.
- D. To segregate broadcast domains.

Answer: B

QUESTION NO: 136

In OSPF, how is the Router ID determined?

- A. By the highest IP address on any of the router's active interfaces.
- B. By the lowest MAC address of the router.
- C. By a manually configured value in the router configuration.
- D. By the highest loopback interface IP address.

Answer: D

QUESTION NO: 137

You are creating an IP fabric underlay and want to use OSPF as your routing protocol.

In this scenario, which statement is correct?

- A. All leaf devices must be configured in separate OSPF areas.
- B. All leaf and spine devices must be the same model to ensure the proper load-balancing behavior.
- C. Interface speeds should be the same throughout the fabric to ensure that all links are utilized.
- D. All spine devices must use the same router ID

Answer: C

Explanation:

When creating an IP fabric underlay using OSPF as the routing protocol, consistent interface speeds are important to ensure optimal traffic distribution and utilization of all links. Step-by-Step Breakdown:

OSPF and Interface Speeds:

OSPF calculates the cost of a link based on its bandwidth.

The default cost calculation in OSPF is:

If interface speeds vary significantly, OSPF may choose paths with lower cost (higher bandwidth), resulting in some links being underutilized.

Equal Utilization:

To ensure that all links are equally utilized in an IP fabric, it is recommended to maintain uniform interface speeds across the fabric. This ensures balanced load sharing across all available paths. Juniper Reference: IP Fabric with OSPF: Juniper recommends consistent interface speeds to maintain even traffic distribution and optimal link utilization in IP fabric underlay designs.

QUESTION NO: 138

Which OSPF packet types are used to establish and maintain neighbor relationships?

(Choose two)

- A. Hello packets
- B. Database Description packets
- C. Link State Request packets

D. Link State Update packets

Answer: A B

QUESTION NO: 139

Referring to the exhibit, how many milliseconds will pass before the BGP neighbor is considered dead?

[edit]

```
user@router# show protocols bgp
group internal-peers {
    type internal;
    local-address 192.168.6.5;
    export send-direct;
    bfd-liveness-detection {
        minimum-interval 1000;
    }
    neighbor 192.168.6.4;
    neighbor 192.168.40.4;
}
```

A. 1000ms

B. 9000ms

C. 2000ms

D. 3000ms

Answer: D

QUESTION NO: 140

Why is EVPN often used in conjunction with VXLAN in modern data centers?

A. To enable direct Internet access for applications.

B. For providing Layer 2 connectivity over a Layer 3 fabric.

C. To increase the physical bandwidth of the underlying network.

D. To physically separate different customer environments.

Answer: B

QUESTION NO: 141

Exhibit:

 Exhibit

```
[edit routing-options]
user@router# show
static {
    defaults {
        preference 7;
    }
    route 0.0.0.0/0 {
        next-hop 172.25.20.254;
        qualified-next-hop 172.25.20.200 {
            preference 6;
        }
    }
}
```

Referring to the exhibit, which next hop will be preferred in the routing table?

- A. Next hop IP address 172.25.20.254 will be preferred.
- B. Neither next hop will be preferred.
- C. Next hop IP address 172.25.20.200 will be preferred.
- D. Both next hops will be preferred.

Answer: C

Explanation:

In the exhibit, we see a static route configuration with two possible next hops for the default route (0.0.0.0/0):

next-hop 172.25.20.254 with the default preference of 7.

qualified-next-hop 172.25.20.200 with a preference of 6.

Step-by-Step Breakdown:

Preference Value:

In Junos OS, the preference value is used to determine which route should be preferred in the routing table. The lower the preference value, the higher the priority for the route.

Comparison:

In this case:

The next hop 172.25.20.254 has a preference of 7.

The qualified-next-hop 172.25.20.200 has a preference of 6.

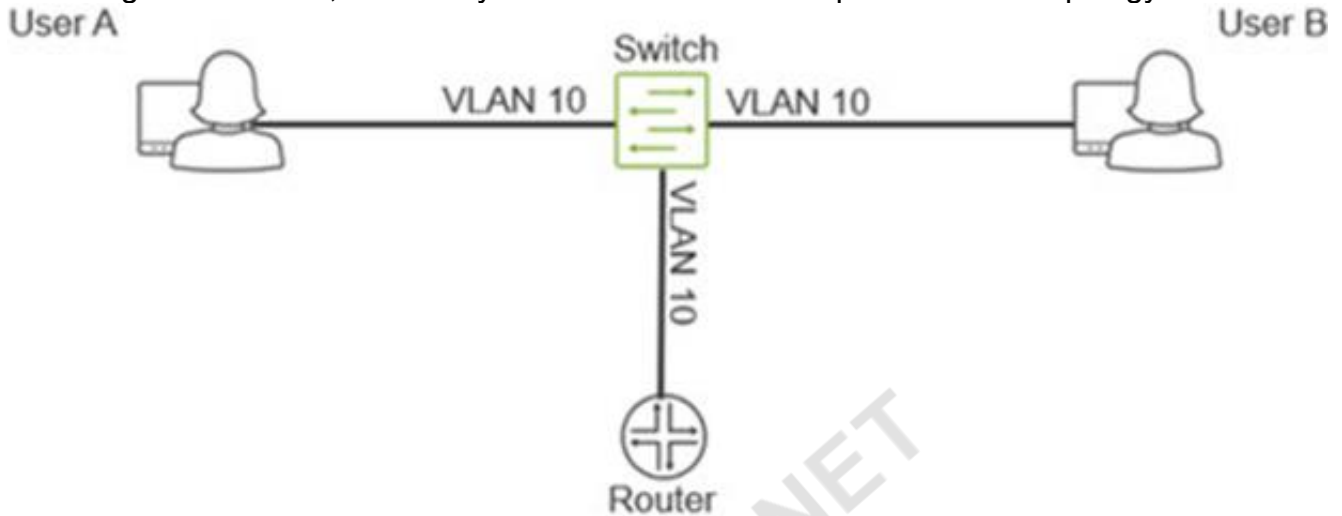
Preferred Next Hop:

Since 172.25.20.200 has a lower preference (6) compared to 172.25.20.254 (7), it will be the preferred next hop in the routing table, assuming both next hops are reachable. Juniper

Reference: Qualified Next Hop: In Junos, static routes with multiple next-hop options are selected based on the preference value, with the lower value being preferred.

QUESTION NO: 142

Referring to the exhibit, how many broadcast domains are present in the topology?



- A. 0
- B. 1
- C. 2
- D. 3

Answer: B

QUESTION NO: 143

Load balancing in routing is used to:

- A. Encrypt data packets.
- B. Distribute traffic evenly across multiple paths.
- C. Compress data to improve transmission speed.
- D. Automatically update routing tables.

Answer: B

QUESTION NO: 144

Which EBGP feature must be used when peering to a non-directly connected peer address?

- A. multihop
- B. AS path
- C. local preference
- D. multipath

Answer: A

QUESTION NO: 145

Which two statements are correct about trunk ports? (Choose two.)

- A. Trunk ports typically connect to other switches and routers.
- B. By default, trunk ports can transmit and receive VLAN tagged traffic.
- C. Trunk ports typically connect to end users and servers.
- D. By default, trunk ports can transmit and receive untagged traffic.

Answer: A B

QUESTION NO: 146

Which of the following statements are correct about the Spine/Leaf architecture? (Choose two)

- A. The Spine layer switches interconnect various Leaf switches.
- B. Each Leaf switch is directly connected to every server in the data center.
- C. The architecture reduces the need for inter-switch link (ISL) redundancy.
- D. Spine switches can connect directly to the data center's external network.

Answer: A D

QUESTION NO: 147

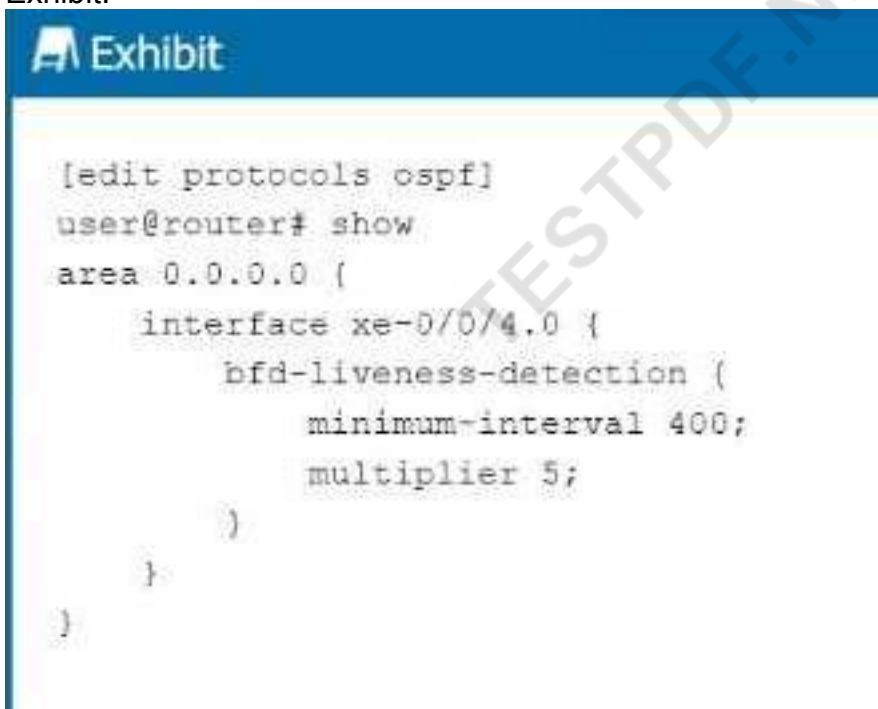
In configuring load balancing on a network router, which aspect is most crucial?

- A. The encryption protocol used
- B. The number of routing tables
- C. The selection of multiple equal-cost paths
- D. The use of aggregate routes

Answer: C

QUESTION NO: 148

Exhibit:



```
[edit protocols ospf]
user@router# show
area 0.0.0.0 {
    interface xe-0/0/4.0 {
        bfd-liveness-detection {
            minimum-interval 400;
            multiplier 5;
        }
    }
}
```

Referring to the exhibit, at which interval will the interface be considered down if no hello packets are received?

- A. 2000 seconds
- B. 400 milliseconds
- C. 400 seconds
- D. 2000 milliseconds

Answer: D

Explanation:

The exhibit shows the configuration of Bidirectional Forwarding Detection (BFD) for OSPF on interface xe-0/0/4.0, with the following parameters: minimum-interval: 400 milliseconds

multiplier: 5 Step-by-Step Breakdown:

BFD Liveness Detection:

BFD is used to detect link failures at sub-second intervals, providing faster convergence times for routing protocols like OSPF. The minimum-interval is the time between BFD control packets (in milliseconds), and the multiplier indicates how many missed BFD packets trigger a failure.

Calculating Failure Detection Time:

The failure detection interval is calculated as:

Failure Interval=minimum-interval×multiplier
 $\text{Failure Interval} = \text{minimum-interval} \times \text{multiplier}$

In this case:
 $400\text{ milliseconds} \times 5 = 2000\text{ milliseconds (2 seconds)}$

Conclusion:
 If no BFD control packets are received within 2000 milliseconds (2 seconds), the interface will be considered down, triggering OSPF to recalculate routes.

Juniper Reference: BFD Configuration: BFD parameters such as minimum-interval and multiplier are used to fine-tune the failure detection time for faster convergence.

QUESTION NO: 149

What type of BGP is used for routing within the same autonomous system?

- A. IBGP
- B. EBGp
- C. Static BGP
- D. Dynamic BGP

Answer: A

QUESTION NO: 150

What does filter-based forwarding in networking typically allow?

- A. Forwarding decisions based on MAC addresses
- B. Load balancing across multiple CPUs
- C. Routing decisions based on extended criteria like packet headers
- D. Aggregation of multiple routes into a single entry

Answer: C

QUESTION NO: 151

Which of the following are functions of MACsec? (Choose two)

- A. It provides Layer 2 encryption.
- B. It prevents MAC flooding attacks.
- C. It offers data integrity and confidentiality.
- D. It enables VLAN tagging.

Answer: A C

QUESTION NO: 152

With regards to graceful restart, which two statements are true? (Choose two.)

- A. A supported routing protocol, such as BGP, must be operational for graceful restart to work.
- B. NSR must be enabled for graceful restart to work.
- C. Dual routing engines are required for graceful restart to work.
- D. The network topology must be stable for graceful restart to work.

Answer: A D

QUESTION NO: 153

In OSPF, what is the function of the Link-State Database (LSDB)?

- A. To store OSPF routing tables.
- B. To maintain a list of OSPF neighbors.
- C. To keep the topology structure of the network.
- D. To log changes in network topology.

Answer: C

QUESTION NO: 154

When configuring static routes, which command is typically used?

- A. set protocols static route
- B. set route aggregate
- C. set firewall filter
- D. set routing-options rib-groups

Answer: A

QUESTION NO: 155

Filter-based forwarding is typically used to:

- A. Increase the security of the routing process.
- B. Implement policy-based routing decisions.
- C. Optimize the routing table size.
- D. Automatically generate routing entries.

Answer: B

QUESTION NO: 156

In the context of Ethernet bridging on Junos devices, what is the purpose of the Spanning Tree Protocol (STP)?

- A. To prevent routing loops.
- B. To provide a mechanism for VLAN tagging.
- C. To prevent Layer 2 switching loops.
- D. To filter MAC addresses.

Answer: C

QUESTION NO: 157

Which two statements are correct about using LACP on a Junos device? (Choose two.)

- A. Both sides of the LACP session must be in active mode to exchange LACP packets on a Junos device.
- B. One side of the LACP session must be in active mode to exchange LACP packets on a Junos device.
- C. LACP is used to monitor the status of an aggregated Ethernet bundle's member interfaces on a Junos device.
- D. LACP is used to automatically add and delete interfaces from aggregated Ethernet bundles on a Junos device.

Answer: B C

QUESTION NO: 158

What does the "qualified-next-hop" feature do when configuring a static route?

- A. It allows you to specify an independent route preference value for an additional next hop.
- B. It specifies the IP address of the directly connected device.
- C. It instructs the device to verify the next-hop is reachable before activating the route.
- D. It instructs the device to look up a static route's next-hop that is not directly connected.

Answer: A

QUESTION NO: 159

In a Virtual Chassis configuration, which of the following is true?

- A. It allows physical stacking of switches that are geographically dispersed.
- B. It reduces the operational complexity of managing multiple switches.
- C. It decreases network redundancy and reliability.
- D. Only two switches can be connected in a Virtual Chassis setup.

Answer: B

QUESTION NO: 160

What is a common issue to check for when troubleshooting a LAG not forming correctly?

- A. Mismatched VLAN configurations
- B. Excessive broadcast traffic
- C. Identical IP addresses on all LAG interfaces
- D. Overutilization of CPU resources

Answer: A

QUESTION NO: 161

In the Junos OS, what is the default behavior of a switch when it receives a frame with an unknown destination MAC address?

- A. It discards the frame.
- B. It floods the frame out of all ports except the one it was received on.
- C. It sends the frame to its default gateway.
- D. It buffers the frame until the MAC address can be learned.

Answer: B

QUESTION NO: 162

How does a layer 2 switch populate its bridge table?

- A. It stores the protocol number of frames traversing the network.
- B. It stores both the source and destination MAC addresses of frames traversing the network.
- C. It stores the destination MAC address of frames traversing the network.
- D. It stores the source MAC address of frames traversing the network.

Answer: D

QUESTION NO: 163

What is the function of the 'Weight' attribute in BGP, and where is it used in the route selection process?

- A. It is used to prioritize routes received from different EBGP peers and is considered before the AS_PATH attribute.
- B. It determines the stability of a route and is checked after the Local Preference attribute.
- C. It is a Cisco-specific attribute used to prefer paths from one BGP speaker over another, evaluated before the Local Preference.
- D. It is used to balance load among multiple links and is considered after the MED attribute.

Answer: C

QUESTION NO: 164

When troubleshooting issues with static routes, what is a key factor to check first?

- A. If the routing table has enough memory
- B. The correctness of the specified next-hop addresses
- C. The speed of the router's CPU
- D. The number of aggregate routes configured

Answer: B

QUESTION NO: 165

In OSPF, which term refers to routers that have interfaces in multiple areas?

- A. Internal Routers
- B. Backbone Routers
- C. Area Border Routers (ABRs)
- D. Autonomous System Boundary Routers (ASBRs)

Answer: C

QUESTION NO: 166

Which of the following VLAN port modes exist in Junos OS? (Choose two)

- A. Tunnel
- B. Access
- C. Trunk
- D. Aggregated

Answer: B C

QUESTION NO: 167

What is the primary difference between IBGP and EBGP?

- A. The way they advertise routes
- B. The administrative distance
- C. The use of AS_PATH attribute
- D. The requirement for a full mesh topology

Answer: B

QUESTION NO: 168

Which type of route is used to represent a summary of multiple smaller networks?

- A. Static route
- B. Aggregate route
- C. Generated route
- D. Dynamic route

Answer: B

QUESTION NO: 169

Under which Junos hierarchy is a static route configured?

- A. system
- B. routing-instances
- C. routing-options
- D. protocols

Answer: C

QUESTION NO: 170

What does MACsec stand for?

- A. MAC Security
- B. Media Access Control Security
- C. MAC Sublayer Encryption
- D. Media Access Control Sublayer Encryption

Answer: B

QUESTION NO: 171

Referring to the exhibit, which protocol is associated with the route selected for traffic destined to 10.255.14.177?

```
user@router> show route
```

```
inet.0: 5 destinations, 7 routes (5 active, 0 holddown, 0 hidden)
```

```
+ = Active Route, - = Last Active, * = Both
```

```
192.168.0.0/30    *[Direct/0] 3d 07:03:01
```

```
> via ge-1/2/0.4
```

```
[BGP/8] 01:04:49, localpref 100
```

```
AS path: 65000 I
```

```
> to 192.168.0.1 via ge-1/2/0.4
```

```
192.168.0.1/32    *[Local/0] 3d 07:03:01
```

```
Local via ge-1/2/0.4
```

```
10.255.14.177/32  *[BGP/8] 01:04:49, localpref 100
```

```
AS path: 65000 I
```

```
> to 192.168.0.1 via ge-1/2/0.4
```

```
[OSPF/10] 3d 07:02:16, metric 1
```

```
> to 192.168.0.1 via ge-1/2/0.4
```

```
10.255.71.24/32   *[Direct/0] 3d 07:03:01
```

```
> via lo0.2
```

```
224.0.0.5/32      *[OSPF/10] 5d 03:42:16, metric 1
```

```
MultiRecv
```

- A. BGP
- B. Direct
- C. OSPF
- D. Local

Answer: A

QUESTION NO: 172

What is the default route preference for a static route on a Junos device?

- A. 130
- B. 15
- C. 10
- D. 5

Answer: D

QUESTION NO: 173

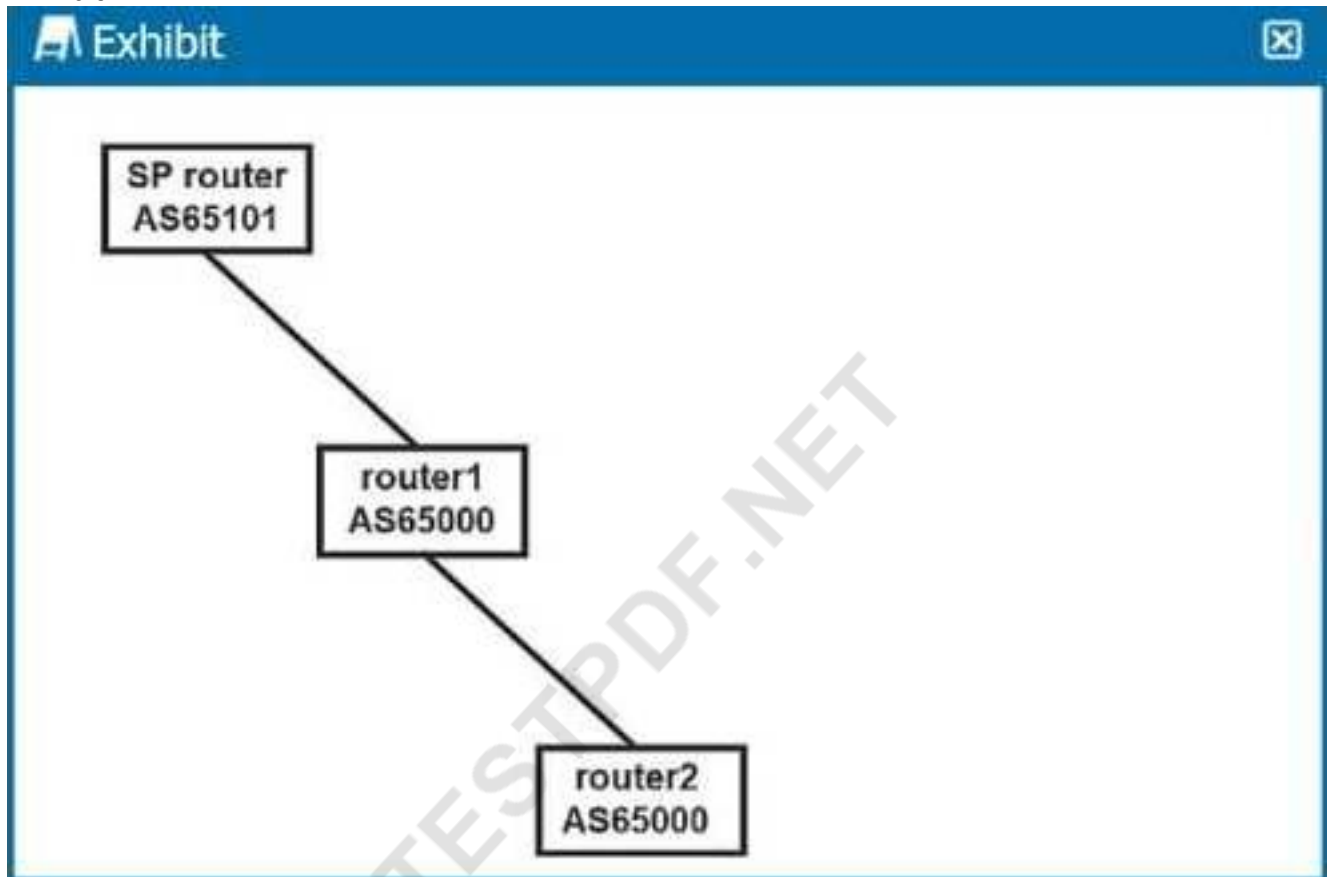
In the context of Layer 2 switching, what does RSTP stand for?

- A. Rapid Spanning Tree Protocol
- B. Routing Switch Transport Protocol
- C. Redundant Switching Tree Protocol
- D. Resilient Switching Transport Protocol

Answer: A

QUESTION NO: 174

Exhibit:



Referring to the exhibit, which two statements are correct about default BGP advertisements? (Choose two.)

- A. When routes advertised by router2 are received by the SP router, they will contain the next-hop address of router2.
- B. When routes advertised by router2 are received by the SP router, they will contain the next-hop address of router1.
- C. When routes advertised by the SP router are received by router2, they will contain the next-hop address of the SP router.
- D. When routes advertised by the SP router are received by router2, they will contain the next-hop address of router1.

Answer: B D

Explanation:

The exhibit shows a BGP peering scenario between three routers: router1 and router2 are part of the same AS (AS65000), while the SP router is in a different AS (AS65101). This

indicates an EBGP (External BGP) peering between the SP router and router1, and IBGP between router1 and router2.

Step-by-Step Breakdown:

Next-Hop Behavior in BGP:

IBGP: In IBGP, the next-hop address is not modified when advertising routes within the same AS. Thus, when router1 advertises routes learned from router2 to the SP router, it will keep the next-hop address of router1, not router2.

EBGP: In EBGP, the next-hop address is modified. When router1 receives routes from the SP router, it will advertise them to router2 with the next-hop address of router1.

Route Propagation:

Routes received by router1 from router2 will be advertised to the SP router with router1 as the next hop.

Similarly, routes advertised by the SP router will be passed on to router2, with router1 remaining as the next hop.

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Reference: BGP Next-Hop: Juniper's BGP implementations follow standard BGP next-hop behavior, where the next-hop is modified in EBGP but not in IBGP, ensuring proper route advertisement across autonomous systems.

QUESTION NO: 175

What is the primary purpose of MAC address filtering on a switch?

- A. To control the flow of encrypted data.
- B. To limit the rate of incoming packets.
- C. To prevent unauthorized devices from accessing the network.
- D. To distribute traffic evenly across multiple links.

Answer: C

QUESTION NO: 176

What is the primary use of a virtual link in OSPF?

- A. To connect OSPF areas to external networks.
- B. To encrypt OSPF traffic.
- C. To connect an OSPF area to the backbone area through a non-backbone area.
- D. To balance load across multiple links.

Answer: C

QUESTION NO: 177

Generated routes in a network are mainly used to:

- A. Reduce the overall size of the routing table.
- B. Manually configure specific network routes.
- C. Automatically create a route when a specified condition is met.
- D. Summarize external routes into the network.

Answer: C

QUESTION NO: 178

Fill in the blank: In BGP, the _____ attribute is used to influence the best path selection process.

- A. MED (Metric or Multi-Exit Discriminator)
- B. AS_PATH
- C. Local Preference
- D. Community

Answer: A

QUESTION NO: 179

Which of the following is an important consideration when configuring BFD?

- A. Setting a very high multiplier to avoid false positives
- B. Using the same BFD interval on all devices in the network
- C. Adjusting the BFD interval and multiplier according to network needs
- D. Disabling all other failure detection mechanisms

Answer: C

QUESTION NO: 180

What is the primary role of an IRB interface in a VLAN configuration?

- A. To provide a gateway for VLAN traffic.
- B. To encrypt VLAN traffic.
- C. To prioritize VLAN traffic.
- D. To filter VLAN traffic based on MAC addresses.

Answer: A

QUESTION NO: 181

What are two common reasons for BGP routes to be in the hidden state? (Choose two.)

- A. Routes are being rejected by a policy
- B. Routes are being filtered from the inet.2 routing table
- C. The next hop is unresolvable
- D. The BGP neighbor is in a different autonomous system

Answer: A C

QUESTION NO: 182

Which OSPF router type connects an OSPF area to non-OSPF networks?

- A. Area Border Router (ABR)
- B. Autonomous System Boundary Router (ASBR)
- C. Internal Router
- D. Backbone Router

Answer: B

QUESTION NO: 183

Which statement is correct about a five-stage IP fabric topology?

- A. There is a maximum of five hops between any end-devices

- B. They are commonly referred to as interconnected data centers
- C. There is a maximum of three hops between any end-devices
- D. They are typically deployed in small to medium data centers

Answer: A

QUESTION NO: 184

What is the primary purpose of using static routes in a network?

- A. To automatically adjust routes based on network traffic
- B. To provide a manually configured routing entry
- C. To aggregate multiple routes into one
- D. To generate routing entries dynamically

Answer: B

QUESTION NO: 185

Referring to the exhibit, why are the BGP routes hidden?

```

user@host> show route hidden
inet.0: 25 destinations, 26 routes (24 active, 0 holddown, 1 hidden)
Restart Complete
+ = Active Route, - = Last Active, * = Both
127.0.0.1/32      [Direct/0] 04:26:38
                  > via lo0.0

privatel__inet.0: 2 destinations, 3 routes (2 active, 0 holddown, 0 hidden)
red.inet.0: 6 destinations, 8 routes (4 active, 0 holddown, 3 hidden)
Restart Complete
+ = Active Route, - = Last Active, * = Both
10.5.5.5/32      [BGP/170] 03:44:10, localpref 100, from 10.4.4.4
                  AS path: 100 I
                  Unusable
10.12.1.0/24     [BGP/170] 03:44:10, localpref 100, from 10.4.4.4
                  AS path: 100 I
                  Unusable
10.12.80.4/30    [BGP/170] 03:44:10, localpref 100, from 10.4.4.4
                  AS path: I
                  Unusable

```

- A. Load balancing is not enabled.
- B. There are too many hops to the destination.
- C. The BGP next hop is unreachable.
- D. Other routes are selected because of better metrics.

Answer: C

Explanation:

In the exhibit, the BGP routes are marked as hidden. This typically happens when the routes are not considered valid for use, but they remain in the routing table for reference. One common reason for BGP routes being hidden is that the next hop for these routes is

unreachable.

Step-by-Step Breakdown:

BGP Next Hop:

In BGP, when a route is received from a neighbor, the next hop is the IP address that must be reachable for the route to be used. If the next hop is unreachable (i.e., the router cannot find a path to the next-hop IP), the route is marked as hidden. Analyzing the Exhibit: The exhibit shows that the BGP next hop for all hidden routes is 10.4.4.4. If this IP is unreachable, the BGP routes from that neighbor will not be considered valid, even though they appear in the routing table.

Verification:

Use the command `show route 10.4.4.4` to check if the next-hop IP is reachable.

If the next-hop is not reachable, the BGP routes will be hidden. Resolving the next-hop reachability issue (e.g., fixing an IGP route or an interface) will allow the BGP routes to become active. Juniper Reference: Junos Command: `show route hidden` displays routes that are not considered for forwarding. Troubleshooting: Check the next hop reachability for hidden BGP routes using `show route <next-hop>`.

QUESTION NO: 186

Which of the following are true about Layer 2 and Layer 3 strategies in data center architectures? (Choose two)

- A. Layer 2 strategies are typically used for creating large, flat networks.
- B. Layer 3 strategies cannot be used in conjunction with Layer 2 environments.
- C. Layer 3 strategies help in reducing broadcast domains.
- D. Layer 2 strategies are primarily used for inter-data center connectivity.

Answer: A C

QUESTION NO: 187

In Junos, which command is used to display the status of all interfaces?

- A. `show interfaces terse`
- B. `show vlan brief`
- C. `show interfaces descriptions`
- D. `show ethernet-switching table`

Answer: A

QUESTION NO: 188

In a traditional multi-tier architecture, which layer is responsible for aggregating access layers and providing connectivity to the core layer?

- A. Core Layer
- B. Distribution Layer
- C. Access Layer
- D. Edge Layer

Answer: B

QUESTION NO: 189

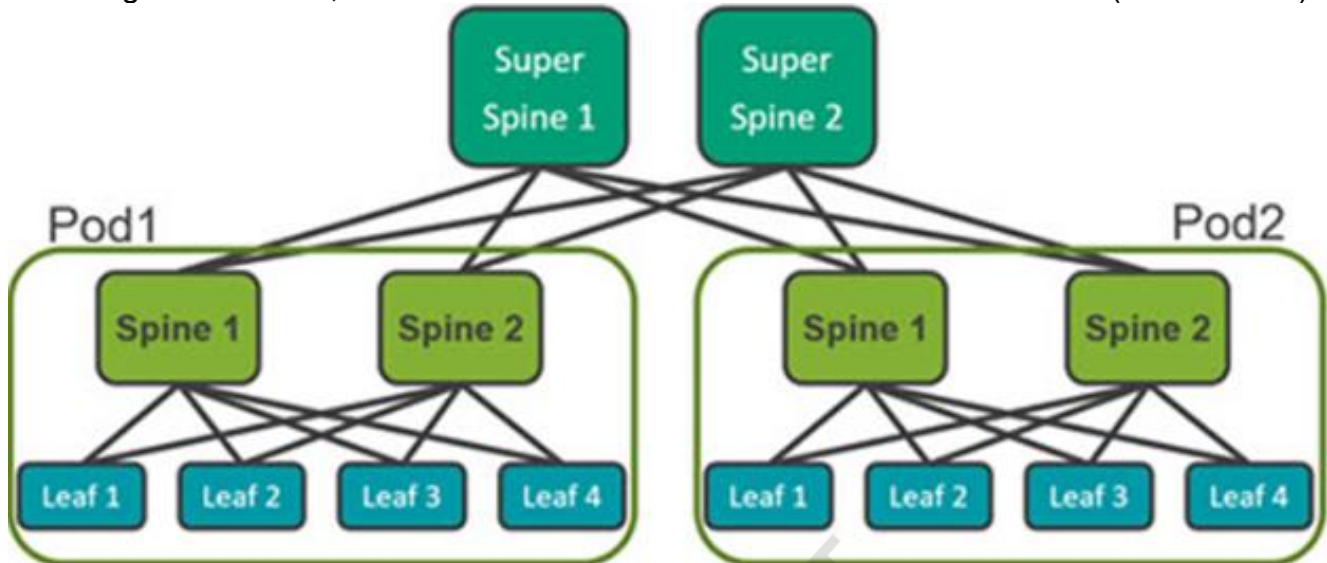
What is the primary purpose of OSPF in network routing?

- A. To redistribute routes between autonomous systems.
- B. To provide path vector routing within an autonomous system.
- C. To offer dynamic routing within an autonomous system using link-state information.
- D. To encrypt data traffic between different networks.

Answer: C

QUESTION NO: 190

Referring to the exhibit, which two statements are correct about the IP fabric? (Choose two.)



- A. Only traffic between pods must traverse the super spine devices.
- B. All traffic must traverse the super spine devices.
- C. This is a three-stage IP fabric.
- D. This is a five-stage fabric.

Answer: A D

QUESTION NO: 191

In the context of protocol-independent routing, what is the main difference between static and generated routes?

- A. Static routes require manual configuration, while generated routes are created dynamically.
- B. Static routes are used for load balancing, while generated routes are not.
- C. Generated routes support summarization, while static routes do not.
- D. Generated routes are manually configured, while static routes are automatically created.

Answer: A

QUESTION NO: 192

Referring to the exhibit, you notice that after committing the configuration, the ae0 and ae1 interfaces appear in a link down state.

Exhibit

```
[edit]
user@switch# show chassis
aggregated-devices {
  ethernet {
    device-count 2;
  }
}
[edit]
user@switch# run show interfaces terse | match ae
ae0                up    down
ae1                up    down
```

Which statement is correct in this scenario?

- A. No operational interfaces have been added to the LAG interfaces.
- B. No traffic is traversing the LAG interfaces.
- C. The LAG interfaces are in a passive state.
- D. The LAG interfaces are in aggressive mode.

Answer: A

Explanation:

In the exhibit, the ae0 and ae1 interfaces are in a link down state. This occurs when no physical interfaces (member interfaces) have been added to the LAG (Link Aggregation Group) interfaces, or the member interfaces are not operational.

Step-by-Step Breakdown:

LAG Configuration:

A LAG interface (aggregated Ethernet interface) is a logical interface that combines multiple physical interfaces for redundancy and increased bandwidth. The LAG will only be operational if at least one member interface is active and configured correctly.

No Operational Member Interfaces:

If no member interfaces are added or if the member interfaces are down, the LAG will remain in a down state, as shown in the exhibit for ae0 and ae1. Resolution:

Verify that physical interfaces have been added to the LAG using commands like:

LAG Interface Status: In Juniper, the link status of the LAG depends on its member interfaces, which must be operational for the LAG to function.

QUESTION NO: 193

Exhibit:

 Exhibit

```
[edit]
user@router# show protocols
bgp {
  group 1 {
    local-as 65101;
  }
  neighbor 172.16.1.1 {
    peer-as 65201;
  }
}
[edit]
user@router# show routing-options
router-id 192.168.100.1;
autonomous-system 65000;
```

Referring to the exhibit, which statement is correct?

- A. The configuration will commit successfully and BGP group 1 will operate as IBGP.
- B. The configuration will commit successfully and BGP group 1 will operate as EBGP.
- C. BGP group 1 requires a type external parameter.
- D. BGP group 1 requires a type internal parameter.

Answer: B

Explanation:

In the exhibit, BGP is configured with local AS 65101 and a neighbor at 172.16.1.1 in peer AS 65201.

This setup involves two different Autonomous Systems (AS), indicating an External BGP (EBGP) configuration.

Step-by-Step Breakdown:

EBGP vs. IBGP:

EBGP is used between routers in different ASes. In this case, the local AS is 65101 and the peer AS is 65201, meaning the BGP session is EBGP.

IBGP is used between routers within the same AS, which is not applicable here as the AS numbers are different.

BGP Group Configuration:

The configuration does not require a type external parameter because Junos OS automatically recognizes the session as EBGP when the local and peer AS numbers are different. The BGP session will operate as EBGP, and the configuration will commit

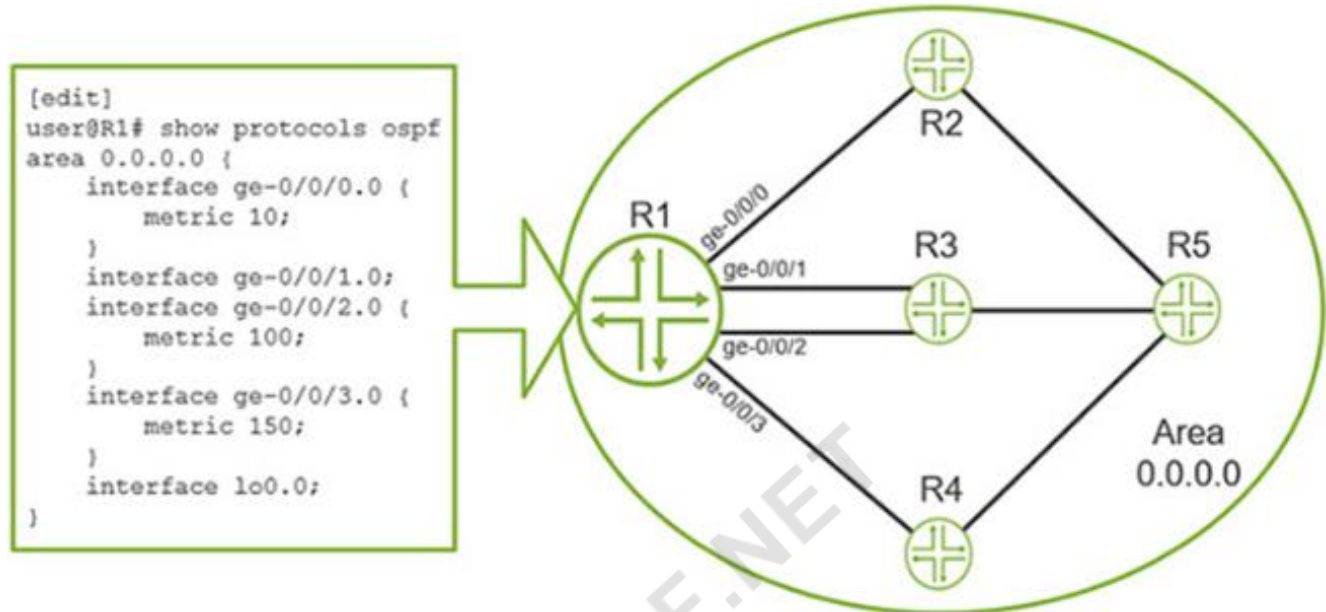
successfully.

Juniper

Reference: BGP Configuration: In Juniper, EBGP is automatically recognized when the local and peer AS numbers differ, without needing to specify type external.

QUESTION NO: 194

You are configuring OSPF in your network as shown in the exhibit. R1 has been configured with interface metrics. All other links in the network are using the default interface metric. In this scenario, which OSPF interface on R1 will be preferred when routing traffic between R1 and R5?



- A. ge-0/0/0
- B. ge-0/0/3
- C. ge-0/0/1
- D. ge-0/0/2

Answer: C

QUESTION NO: 195

Which statement is correct about OSPF?

- A. OSPF is link-state protocol
- B. OSPF is a path-vector protocol
- C. OSPF is a distance-vector protocol
- D. OSPF is a device-state protocol

Answer: A

QUESTION NO: 196

In a Spine/Leaf IP-Fabric architecture, what is the primary role of the leaf switches?

- A. To provide direct connectivity to servers.
- B. To interconnect different data centers.
- C. To perform deep packet inspection.

D. To manage the overall data center network.

Answer: A

QUESTION NO: 197

Fill in the blank: In BGP, a route received from an _____ peer will not be advertised to another _____ peer by default.

A. IBGP, IBGP

B. EBGP, EBGP

C. EBGP, IBGP

D. Static, Dynamic

Answer: A

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